

Economic Consequences of Sequence-Dependent Betting and Its Psychological Underpinnings

ONLINE APPENDIX

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A EXPECTED UTILITY FRAMEWORK

A.1 Risk Preference Estimation

Individual risk preferences are estimated from the Holt–Laury (HL) task using a Bayesian choice model with fixed, informative priors on participant-level parameters. Each participant i is assigned an independent CRRA risk-aversion parameter r_i and choice-precision parameter λ_i , with

$$r_i \sim \mathcal{N}(0.27, 0.36), \quad \lambda_i \sim \text{Lognormal}(\log 30, 0.5),$$

following the calibration of Bland (2023). For each HL row t , expected utility under CRRA is computed for both lotteries,

$$U_A = p_t u(A_1; r_i) + (1 - p_t) u(A_2; r_i), \quad U_B = p_t u(B_1; r_i) + (1 - p_t) u(B_2; r_i),$$

and choices follow a logistic link scaled by individual precision,

$$y_t \sim \text{Bernoulli}(\text{logistic}[\lambda_i(U_A - U_B)]).$$

Posterior draws $r_i^{(m)}$ form the basis for all quantities defined below.

Preregistration Deviations. (Points 1–4 were changes made in the post-pilot addendum to the preregistration — see Appendix C.2 of the preregistration. Point 5 was a change made after observing the main data.)

1. The preregistered HL task used a standard 10-item list; following the pilot, the list was extended to 20 items with probability steps of 0.05 to improve identification of r_i .

2. A response-consistency indicator is defined as

$$\text{HL-inconsistent}_i = \begin{cases} 1 & \text{if participant } i \text{ switches from the risky to the safe} \\ & \text{lottery in a later row,} \\ 0 & \text{otherwise.} \end{cases}$$

3. The model is fitted separately for the FN and FP samples and on the pooled sample as a diagnostic; the correlation between treatment-specific and pooled estimates of r_i is reported in Appendix X.
4. The preregistration referred to the described procedure as a “hierarchical Bayesian model”; this was a labeling error. The estimation procedure, priors, and likelihood are unchanged; only the terminology is corrected.
5. Main data analysis revealed that Bayesian r_i estimates for HL-inconsistent participants diverge substantially from classical switchpoint-based estimates and would distort EU-optimal stake computations. Confirmatory analyses involving r_i are therefore restricted to HL-consistent participants ($\text{HL-inconsistent}_i = 0$); full-sample results are reported as a robustness check in Appendix X.

A.2 CRRA Utility, Optimal Stakes, and Welfare Loss

For stake a from endowment $e = 100$ ECU with multiplier m , the **expected utility of the stake** is:

$$EU(a; r) = \frac{1}{2} u(e - a + ma; r) + \frac{1}{2} u(e - a; r).$$

Preferences follow the constant relative risk aversion (CRRA) utility function

$$u(x; r) = \begin{cases} \frac{x^{1-r}}{1-r}, & r \neq 1, \\ \ln x, & r = 1, \end{cases} \quad x > 0.$$

The CRRA specification requires strictly positive wealth. On the integer stake grid

$a \in \{1, \dots, e\}$, the only payoff state with non-positive wealth is the losing outcome of an all-in stake ($a = e$), which yields $e - a = 0$; at this point $u(\cdot; r)$ diverges to $-\infty$ for $r \geq 1$. To keep expected utility finite for every posterior draw of r_i , this state is evaluated at a small positive floor $x_{\min} = 0.01$ ECU. The floor is a numerical regularization of the utility argument, not an adjustment to the stake or to any realized payoff: the all-in stake and its zero-wealth losing outcome are preserved, and only the evaluation of $u(0; r)$ is replaced. Because $x_{\min}$ enters the welfare metric in RQ3 through the certainty equivalent of all-in losing trials, we verify that the reported welfare loss is insensitive to its value; this robustness check is reported in Appendix [App:...](#)

The **normatively optimal stake** is the integer maximiser of expected utility over the feasible action space,

$$a^*(r_i) = \arg \max_{a \in \{0, 1, \dots, e\}} EU(a; r_i),$$

where $a = 0$ denotes abstention and $a = e$ staking the entire endowment, and ties are resolved in favour of the smaller stake. This discrete maximisation is the operative definition of a^* for every posterior draw of r_i : it is exact on the integer choice set, applies uniformly across risk-loving ($r < 0$), risk-neutral ($r = 0$), and risk-averse ($r > 0$) preferences, and requires no appeal to interior or boundary case distinctions.¹

The **certainty equivalent** of a stake a is the sure payoff $c(a; r) > 0$ that the participant

¹For $r > 0$, $r \neq 1$, the continuous relaxation admits the closed-form interior optimum

$$\tilde{a}(r) = e \frac{(m-1)^{1/r} - 1}{(m-1)^{1/r} + (m-1)}, \quad \tilde{a}(1) = e \frac{m-2}{2(m-1)},$$

which makes the comparative statics transparent: the optimal stake share $\tilde{a}(r)/e$ is increasing in the multiplier m and decreasing in risk aversion r , and is strictly positive only when $m > 2$. This is what underlies the treatment contrast. Under the favourable multiplier ($m = 2.5$) the interior optimum is positive and economically large for all but extreme risk aversion, making a positive stake normatively optimal; under the unfavourable multiplier ($m = 1.9 < 2$) the unconstrained optimum is negative, so the constrained optimum is the corner $a^* = 0$ for all but sufficiently risk-loving participants. When the optimum is interior, it lies at one of the integers bracketing $\tilde{a}(r)$; at the corners, and across the regions where the closed form is singular or undefined ($r \leq 0$, $r \rightarrow 0$, $r \rightarrow 1$), the optimum is recovered directly by the discrete maximisation. All reported values of $a^*(r_i)$ are computed by that discrete maximisation, which is exact on the integer choice set and requires no appeal to interiority; the closed form is used only to characterise the solution and its comparative statics.

values equally to the bet, defined implicitly by

$$u(c(a; r); r) = EU(a; r).$$

Because $u(\cdot; r)$ is continuous and strictly increasing on $(0, \infty)$, this equation has a unique positive solution, obtained by inverting the utility function:

$$c(a; r) = \begin{cases} ((1 - r) EU(a; r))^{1/(1-r)}, & r \neq 1, \\ \exp(EU(a; r)), & r = 1. \end{cases}$$

The closed-form inverse is used whenever it is numerically admissible, namely when $(1 - r) EU(a; r) > 0$ and the result is finite and strictly positive. In the remaining cases—including risk parameters near the singularities $r \rightarrow 0$ and $r \rightarrow 1$, where the exponent $1/(1 - r)$ becomes numerically unstable—the certainty equivalent is computed as the unique positive root of $u(c; r) = EU(a; r)$ by bracketed root-finding. The two procedures coincide wherever both apply, so this is a numerical safeguard rather than a change of definition.

The money-metric **welfare loss** from choosing stake a_{is} rather than the expected-utility-optimal stake a_{is}^* is the difference in certainty equivalents,

$$\Delta c_{is} = c(a_{is}^*; r_i) - c(a_{is}; r_i),$$

expressed in ECU and interpretable as the sure amount the participant forgoes by deviating from the optimum. Since a_{is}^* maximises expected utility and the certainty equivalent is strictly increasing in expected utility, $\Delta c_{is} \geq 0$ by construction. Strictly negative values can arise only from numerical error—when a_{is} is adjacent to a_{is}^* and posterior draws of r_i induce small inversion inaccuracies—and are set to zero prior to estimation, so that Δc_{is} is a non-negative welfare loss throughout.

Preregistration Deviations

1. The preregistration specified that a^* be computed from the closed-form interior optimum (rounded to the nearest integer ECU) where numerically stable, with discrete maximisation as a fallback. The implementation instead uses the discrete maximisation over $\{0, 1, \dots, e\}$ throughout. This yields the identical estimand while avoiding the nearest-integer approximation, which is not guaranteed to coincide with the integer maximiser of expected utility; the closed-form expressions are retained for analytical characterisation only.
2. The preregistration specified that posterior uncertainty in the Holt–Laury risk parameters r_i be propagated to the derived quantities $a^*(r_i)$ and the certainty equivalents by recomputing them at every iteration *within* the sequence-level models—in effect, estimating the risk and sequence-level models jointly. Pilot analysis showed this joint estimation to be computationally prohibitive, and the post-pilot addendum (Preregistration, Appendix C.3) decoupled the two stages, computing the r_i -dependent quantities from the Holt–Laury posterior draws outside the sequence-level models and summarising them to participant-level benchmarks before estimation. The implemented analysis follows this decoupling and, for the welfare-loss analysis, retains the full draw distribution rather than a prior summary, so that more of the r_i posterior is carried forward than the addendum required. Concretely, optimal stakes are computed at each posterior draw, $a^*(r_i^{(k)})$, by discrete maximisation over $\{0, 1, \dots, e\}$ and cached as the full draw distribution. For the welfare-loss analysis (RQ3), the certainty-equivalent loss is evaluated draw-wise, $\Delta c_{is}^{(k)} = c(a_{is}^*; r_i^{(k)}) - c(a_{is}; r_i^{(k)})$, and the trial-level outcome entering the model is its posterior mean, $y_{is} = E_k[\max(\Delta c_{is}^{(k)}, 0)]/e$; the nonlinear certainty-equivalent mapping makes this draw-wise evaluation informative. For the stake-calibration analysis (RQ2), where a^* enters linearly as the benchmark in the deviation $\Delta a_{is} = a_{is} - a_{is}^*$, its posterior mean $\tilde{a}_i^* = E_k[a^*(r_i^{(k)})]$ is a sufficient benchmark and is used directly. The decoupling does not affect identification of the behavioural estimands: under the sequential procedure r_i is identified solely from the risk-elicitation task and enters as a fixed, independently measured benchmark, against which betting is tested but by which it is not revised. The RQ2 and RQ3 estimands are therefore

unchanged; the deviation concerns only the estimation route.

B DETAILS ON EMPIRICAL STRATEGY

B.1 Posterior Classification Conventions

Sequence- and participant-level posterior summaries are reported with descriptive labels based on posterior concentration outside the practical equivalence region. These labels are purely expository and do not function as inferential criteria; all substantive conclusions rely on the full posterior distributions.

Table B1: Descriptive classification labels for sequence-level posterior summaries.

	Quantity	Estimand	Direction	Strong	Moderate	Weak	Neutral
RQ1	$U_s^b(\cdot)$	μ_s^b	Underbetting	$[0.95, 1]$	$[0.80, 0.95)$	$[0.50, 0.80)$	< 0.50
RQ2	$U_s^a(\cdot)$	μ_s^a	Under-staking	$[0.95, 1]$	$[0.80, 0.95)$	$[0.50, 0.80)$	< 0.50
	$O_s^a(\cdot)$		Over-staking	$[0.95, 1]$	$[0.80, 0.95)$	$[0.50, 0.80)$	< 0.50
RQ3	$L_s(\cdot)$	μ_s^c	Welfare loss	$[0.95, 1]$	$[0.80, 0.95)$	$[0.50, 0.80)$	< 0.50
RQ4	$H_s(\cdot)$	μ_s^h	Head-bias	$[0.95, 1]$	$[0.80, 0.95)$	$[0.50, 0.80)$	< 0.50
	$T_s(\cdot)$		Tail-bias	$[0.95, 1]$	$[0.80, 0.95)$	$[0.50, 0.80)$	< 0.50

For RQ1, if overall betting rates are substantially below the normative benchmark, $U_s^b(\cdot)$ may approach one for most sequences; interpretation then relies primarily on posterior medians and credible intervals of μ_s^b .

For RQ4, sequences with both $H_s(\cdot)$ and $T_s(\cdot)$ below 0.50 are classified as neutral; $P(|\mu_s^h - 0.5| > \delta)$ is additionally reported, capturing absolute deviations from parity.

Table B2: Descriptive classification labels for participant-level posterior summaries.

	Quantity	Estimand	Direction	Solid	Likely	Leaning	Neutral
RQ1	$U_i^b(\cdot)$	μ_i^b	Underbetter	$[0.95, 1]$	$[0.90, 0.95)$	$[0.75, 0.90)$	< 0.75
RQ2	$U_i^a(\cdot)$	μ_i^a	Under-staker	$[0.95, 1]$	$[0.90, 0.95)$	$[0.75, 0.90)$	< 0.75
	$O_i^a(\cdot)$		Over-staker	$[0.95, 1]$	$[0.90, 0.95)$	$[0.75, 0.90)$	< 0.75
RQ3	$L_i(\cdot)$	μ_i^c	Welfare loser	$[0.95, 1]$	$[0.90, 0.95)$	$[0.75, 0.90)$	< 0.75
RQ4	$H_i(\cdot)$	μ_i^h	Headish	$[0.95, 1]$	$[0.90, 0.95)$	$[0.75, 0.90)$	< 0.75
	$T_i(\cdot)$		Tailish	$[0.95, 1]$	$[0.90, 0.95)$	$[0.75, 0.90)$	< 0.75

For RQ4, participants with both $H_i(\cdot)$ and $T_i(\cdot)$ below 0.75 are classified as neutral; $P(|\mu_i^h - 0.5| > \delta)$ is additionally reported, capturing absolute deviations from parity.

B.2 Preregistered EX1 and EX4: Framework and Consequences of GHI Collapse

The preregistered EX1 analysis comprised three parts. Part I — the anchor-based distributional similarity framework and the Gamblerish–Hot-handish Index (GHI) — is described in full in Section ?? of the main text. Parts II and III, specified in Sections ?? and ?? below, were estimated as preregistered but yielded no meaningful inferential content: the pure sequences turned out behaviorally neutral, collapsing the GHI toward zero for all participants. The preregistered EX4 specification was similarly affected, as documented in Section ?? below.

Figure ?? illustrates the collapse directly. In the FN treatment, rather than anchoring the extremes of the similarity space, $\bar{H}$ and $\bar{T}$ sit in the interior of the $(w_{\bar{H}}, w_{\bar{T}})$ distribution, surrounded by impure sequences. In the FP treatment the problem is more severe: the entire weight distribution collapses toward 1/3 for all sequences. The root cause is behavioral: participants treat even the most extreme streaks as uninformative, responding to them no differently than to a sequence with no discernible pattern. Consequently $\chi_s \approx 0$ for all sequences, which also renders the preregistered EX4 comparison between anchor-based and canonical rule-based classifications uninformative. EX4 therefore proceeds against μ_s^h from RQ4 directly, as described in Section ??.

B.2.1 EX1–II. Predictors of the GHI

The second part specified a regression of the GHI on dispositional optimism, response time, and risk preferences. Because χ_i is a derived posterior quantity, uncertainty was to be propagated by estimating, for each posterior draw t , a Bayesian regression

$$\chi_i^{(t)} = \alpha + \beta_{\text{opt}} Z_i^{\text{opt}} + \beta_{\text{rt}} Z_i^{\text{rt}} + \beta_r Z_i^r + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma_\varepsilon), \quad (1)$$

where Z_i^{opt} , Z_i^{rt} , and Z_i^r are standardized LOT-R score, log response time, and risk parameter, respectively, with weakly informative priors $\alpha, \beta_{\text{opt}}, \beta_{\text{rt}}, \beta_r \sim N(0, 1)$ and $\sigma_\varepsilon \sim \text{HalfNormal}(0, 1)$.

B.2.2 EX1-III. Pure-Sequence Approximation

The third part specified a comparison between a simple pure-sequence score and the full GHI. For each participant i , the pure-sequence analogue was defined as

$$\chi_i^{\text{pure}} = \mathbf{1}\{h_{i,\bar{H}} = 1\} - \mathbf{1}\{h_{i,\bar{T}} = 1\} \in \{-1, 0, 1\}, \quad (2)$$

where $\chi_i^{\text{pure}} = 1$ indicates hot-handish responses on both pure sequences, $\chi_i^{\text{pure}} = -1$ gamblerish responses on both, and $\chi_i^{\text{pure}} = 0$ mixed responding. Association with the full GHI was to be summarized via the posterior distribution of $\rho_{\text{pure},\chi} = \text{cor}(\chi_i^{\text{pure}}, \chi_i)$, with posterior medians, 95% credible intervals, and $P(\rho_{\text{pure},\chi} > 0)$ reported, alongside posterior distributions of χ_i separately for the three pure-sequence categories $\chi_i^{\text{pure}} \in \{-1, 0, 1\}$.

B.2.3 EX4 GHI vs. Canonical Rules

The preregistered EX4 analysis specified a comparison of canonical rule-based sequence scores $Q_s^{(k)}$ to anchor-based directional tendencies χ_s from EX1, with full posterior uncertainty propagation. For each rule k and posterior draw t , sequence-level directional agreement was defined as

$$\text{Agree}_{\text{seq}}^{(k,t)} = \frac{1}{64} \sum_s \mathbf{1}\{\text{sign}(Q_s^{(k)}) = \text{sign}(\chi_s^{(t)})\}, \quad (3)$$

and association was to be summarized by the posterior distribution of the Spearman correlation

$$\rho_{\text{seq}}^{(k,t)} = \text{cor}_{\text{Spearman}}(Q_s^{(k)}, \chi_s^{(t)}), \quad (4)$$

computed across the 64 sequences, with posterior medians, 95% credible intervals, and $P(\rho > 0)$ reported for each rule k , alongside an analogous participant-level comparison correlating $Q_i^{(k)}$ against $\chi_i^{(t)}$. Since $\chi_s \approx 0$ for all sequences, these analyses are uninformative and not reported; the adapted EX4 analysis using μ_s^h is described in Section ??.

C SUPPLEMENTAL ANALYSES

C.1 Treatments Comparison

Recall that the FP treatment was preregistered as exploratory and assigned a smaller sample (1:3 ratio relative to FN), reflecting the expectation that abstention is EU-optimal for nearly all participants and that betting rates would be too low to support a full hierarchical analysis. As the descriptive statistics reveal, however, FP participants bet at rates and stakes comparable to those observed in FN — a striking convergence given that the two frames carry opposite normative prescriptions, and one that itself warrants explanation. The contrasts reported below are estimated on the full sample within each treatment, including Holt–Laury-inconsistent participants, who were excluded from the confirmatory analysis. Table D3 provides an overview of the key descriptors and Figure C1 illustrates the full participant-level distributions for both treatments; the FN distribution in particular serves as a visual robustness check on the confirmatory results above. Probably the most intriguing finding is that **welfare loss** is comparable across both treatments with medians of 14.5% (FN) and 15.7% (FP), produced by distributions that are heavily concentrated near zero with a long right tail (median $\Delta = -0.02$, 95% CI $[-0.03, -0.01]$), with only a 1% posterior probability that the two frames differ by more than 5 percentage points of the endowment. Despite the severe overbetting, FP participants thus do not seem to incur meaningfully larger welfare losses than their more “rational” FN counterparts. The Spearman correlation of sequence-level medians is weak ($\rho = 0.34$, 95% CI $[0.17, 0.51]$), suggesting that welfare losses are not consistently driven by the same sequences across treatments—this further corroborates the story that who you are affects your earnings more than what sequence you face. Referring back to Figure ??, raw betting rates and conditional stakes are similarly distributed across treatments, which together account for the comparable welfare losses observed in both frames. Turning to the modelled **betting rate**, for which the raw rate serves as a sanity check, FN participants bet more than FP participants (76.9% vs. 68.2%, median $\Delta = +0.09$, 95% CI $[0.07, 0.10]$, $P(\Delta > 0) = 1.00$). However, the practical magnitude of the aggregate difference is

negligible—the posterior probability of exceeding a 10-percentage-point gap is only 0.04, consistent with the participant-level raw distributions being statistically indistinguishable (KS: $D = 0.15$, $p = 0.23$). At the sequence level, contrasts range from near zero to +0.19, with 34 of 64 sequences showing a credible positive gap ($P(\Delta > 0) \geq 0.95$), and the Spearman correlation of sequence-level medians ($\rho = 0.54$, 95% CI $[0.39, 0.67]$) indicates that whatever drives variation in betting across sequences is largely common to both treatments. For completeness, we also report the **stake deviation**. The FN–FP contrast is large, uniform, and entirely negative (participant-level median $\Delta = -0.26$, 95% CI $[-0.27, -0.26]$, $P(\Delta > 0) = 0.00$), holding for all 64 sequences ($P(|\Delta| > 0.05) = 1.00$ everywhere, sequence range $[-0.31, -0.22]$). This is the mechanical consequence of the EU-optimal stake being virtually zero in the FP treatment—given similar raw stakes and betting rates across frames, FP participants must deviate far more from their near-zero optimum than their FN counterparts. The fact that the model recovers this pattern points to the internal consistency of the data.

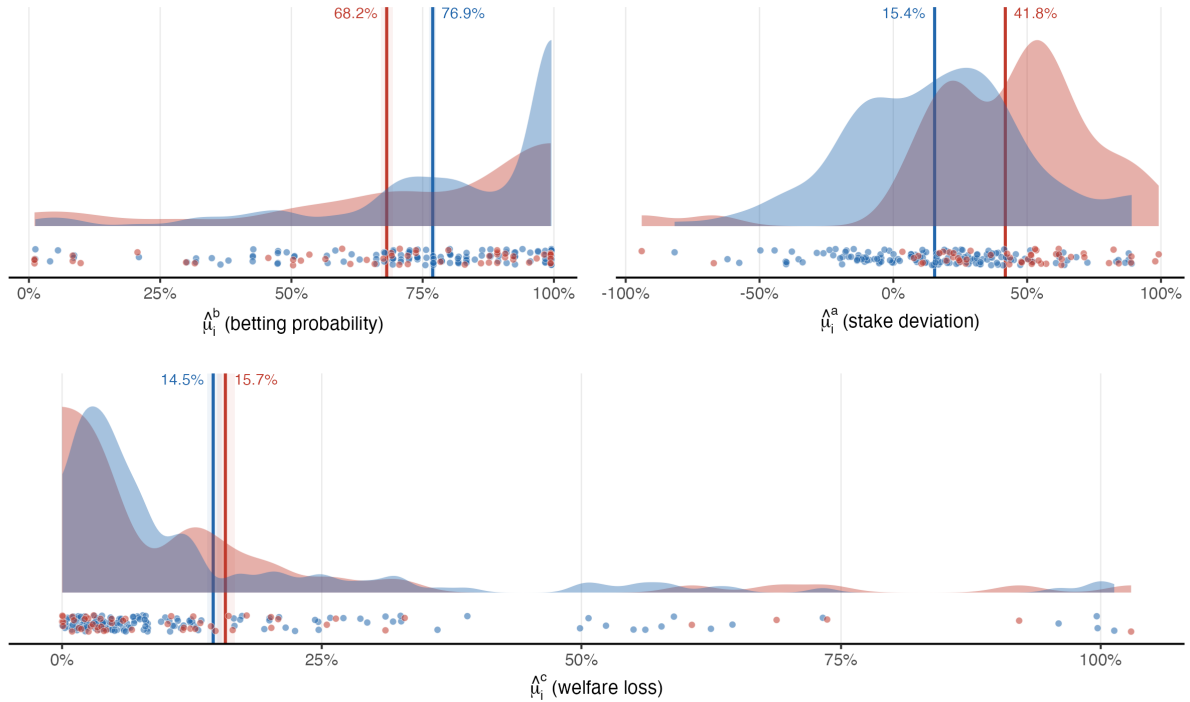


Figure C1

C.2 EX1: GHI collapse

This section documents the collapse of the preregistered EX1 anchor-based framework, referenced in the main text (Section ??). EX1 classifies each sequence’s directional tendency via posterior similarity to two behavioral anchors, the pure sequences HHHHHH (H-pure) and TTTTTT (T-pure), on the grounds that betting with versus against the streak is unambiguously hot-handish or gamblerish only on these two sequences. As reported in the main text, however, both pure sequences are themselves classified as neutral rather than eliciting the expected directional response, undermining their use as anchors. We detail the resulting failure of the anchor-based similarity weights below, and its consequences for the Gamblerish–Hot-handish Index (GHI).

Specifically, each sequence s is assigned normalized similarity weights— $w_{\bar{H}}(s)$, $w_{\bar{T}}(s)$, and $w_0(s)$, summing to one—reflecting its posterior proximity to the H-pure, T-pure, and neutral baseline, respectively. Because the pure anchors are themselves indistinguishable from neutral responding, however, these weights are empirically uninformative. Figure C2 illustrates the problem directly: in the FN treatment, rather than anchoring the extremes of the similarity space, H and T sit in the interior of the $(w_{\bar{H}}, w_{\bar{T}})$ distribution, surrounded by impure sequences. The structured hyperbolic shape of the point cloud may suggest a well-functioning classification, but it is a geometric artifact of the simplex constraint $w_{\bar{H}} + w_{\bar{T}} + w_0 = 1$: any set of three normalized weights projected onto two dimensions must trace out exactly this curve, regardless of whether the underlying similarity scores carry meaningful information. The sequences that end up at the extremes—such as THTTHH ($w_{\bar{H}} \approx 0.97$) or HHTHTT ($w_{\bar{T}} \approx 0.94$)—are not sequences that behaviorally resemble the pure anchors; they are sequences whose Heads-choice probability deviates most from 0.5, placing them far from all three reference points simultaneously. When all similarity scores are near zero, normalization amplifies a negligible difference into an apparently decisive weight. In the FP treatment the problem is more severe: the entire weight distribution collapses toward $1/3$ for all sequences, with virtually no discriminating power across the similarity space. This likely reflects the smaller FP sample, which produces wider posterior distributions, more overlap between sequences, and consequently near-equal similarity

scores that normalization cannot resolve. The root cause is behavioral, not technical: the preregistered framework correctly identifies H-pure and T-pure as the sequences on which hot-hand and gambler’s fallacy responding would be unambiguous, but the data reveal that participants treat even these extreme streaks as uninformative, responding to them no differently than to a sequence with no discernible pattern. Consequently, the directional index returns $\chi \approx 0$ and classifies all sequences as neutral—not because participants show no directional tendency, but because the anchors’ failure to provide informative reference points leaves the classification scheme without useful signal for any sequence.

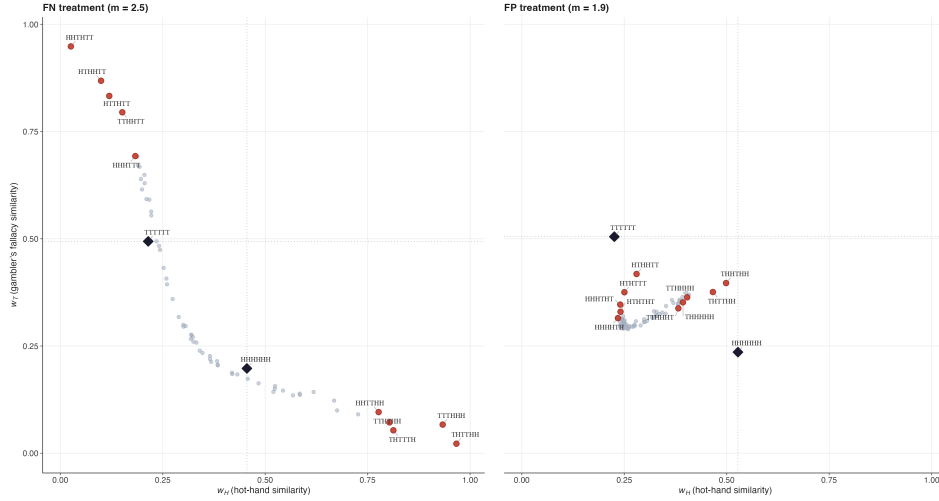


Figure C2

The Gamblerish–Hot-handish Index (GHI) aggregates these sequence-level scores into an individual tendency $\chi_i \in [-1, 1]$, with positive values indicating hot-handishness and negative values gamblerishness—but since all sequence-level scores collapse toward zero, the GHI reduces to an average of uninformative quantities, and the preregistered analyses of its associations with optimism, risk preferences, and response times lack meaningful basis. The preregistered approximation of the full GHI by responses to the pure sequences alone is similarly undermined—ironically, it is precisely the pure sequences whose failure as anchors drives the collapse.

Rather than abandoning the preregistered research question, we pursue it at the level of raw behavior on the pure sequences directly, in the spirit of the preregistered

pure-sequence approximation. Participants are classified as *gambler-consistent* if they bet on a streak reversal on both pure sequences (37.5%/37.0% FN/FP), *hot-hand consistent* if they bet on streak continuation on both (31.6%/30.4%), *inconsistent* if they bet the same side on both regardless of streak direction (16.2%/23.9%), and *unclassified* if they did not place a bet on at least one pure sequence (14.7%/8.7%). Covariate associations are assessed purely descriptively via group means and 95% confidence intervals; the underlying data can be found in the project repository as `ex1_pure_seq_covariates.csv`. The only cells whose intervals exclude zero are z-scored optimism (0.41, 95% CI [0.05, 0.78]) and z-scored log response time (-0.75 , 95% CI $[-1.20, -0.30]$) for the inconsistent group in the FN treatment, who decide faster and score higher on dispositional optimism than the sample average. No covariate credibly distinguishes gambler-consistent from hot-hand consistent participants. The faster response times of the inconsistent group are consistent with random responding—participants not processing streak content deliberate less—and suggest the classification is capturing a meaningful behavioral distinction even if not the one the preregistration envisioned.

C.3 EX4: Canonical Rules

The preregistered comparison between anchor-based directional classifications and canonical rule-based heuristics from the literature cannot be executed as intended, since the EX1 failure renders $\chi_s \approx 0$ for all sequences. We instead correlate the canonical sequence-level scores $Q_s^{(k)}$ directly against the RQ4 estimates μ_s^h (Table C1). All Spearman correlations are near zero and statistically indistinguishable from chance in the FN treatment ($|\rho| \leq 0.09$, all $p > .37$), with no meaningful signal in the FP treatment either once the structural reason is accounted for. The null is not a power issue: the terminal-run structure already identified in RQ4 explains why the canonical rules fail as global predictors. As noted there, sequences ending in a terminal run of at least two are overrepresented among directionally biased sequences (27 of 40) relative to neutral ones (4 of 24), so the run-length rules correctly identify the direction of bias within the biased subset. However, they also flag a substantial fraction of neutral sequences, and since participants bet against the

terminal run on virtually all sequences where a run exists, $Q_s^{(k)}$ is uniformly negative across all applicable sequences—leaving no variance to correlate with μ_s^h . The run-length-2 rule is therefore best understood as a coarse probabilistic heuristic: on sequences ending in a run of at least two, participants tend to bet against the terminal run, consistent with the gambler’s fallacy. Crucially, neither Figure ?? nor its FP counterpart reveals any sequence that credibly elicits the opposite tendency at the population level—directional bias is uniformly gamblerish or neutral across all 64 sequences in both treatments, and hot-handish responding simply does not appear in the data. From a predictive standpoint, the rule is therefore conservative in a useful sense: where it fires, it will more likely than not predict the correct direction of bias, and where it is wrong, the sequence is neutral rather than biased in the opposite direction—the worst-case prediction error is a false positive, not a sign reversal.

Rule	FN treatment ($m = 2.5, n = 147$)			FP treatment ($m = 1.9, n = 50$)		
	$\rho(\mu_s^h)$	$\rho(H_\delta)$	$\rho(T_\delta)$	$\rho(\mu_s^h)$	$\rho(H_\delta)$	$\rho(T_\delta)$
Last outcome	−0.02	−0.04	0.02	−0.21	−0.21	0.21
Run ≥ 2	−0.09	−0.16	0.05	−0.22	−0.22	0.22
Run ≥ 3	−0.07	−0.15	0.01	−0.53*	−0.53*	0.52*
Imbalance	0.08	0.07	−0.10	−0.01	−0.02	0.03
Alternation	0.04	0.04	−0.04	−0.07	−0.07	0.09

Table C1: Spearman correlations between canonical rule-based sequence scores $Q_s^{(k)}$ and RQ4 sequence-level directional estimates: posterior median Heads-choice probability (μ_s^h), posterior probability of Head-bias ($H_{\delta,05}$), and posterior probability of Tail-bias ($T_{\delta,05}$). All 64 sequences included. * $p < .05$; all other correlations $p > .09$. The single significant result in the FP treatment reflects the global dominance of gamblerish responding rather than meaningful rule-based discrimination; see text.

C.4 Hierarchical Joint Model of Betting and Side Choice

Model. To formalize the relationship between the run rule and the imbalance rule, and to test whether they reflect a shared individual disposition or two separable processes, we fit a joint hierarchical Bayesian model over both decision margins simultaneously: whether a participant bets at all (the extensive margin) and, conditional on betting, which side they choose. Each participant is assigned a single, correlated set of random effects spanning both equations, so that any relationship between engagement and directional choice must be recovered from the data rather than assumed by construction.

Let $i \in \{1, \dots, P\}$ index participants and $s \in \{1, \dots, S\}$ index sequences. Extensive-margin observations are indexed $n = 1, \dots, N$ (all trials); side-choice observations are indexed $m = 1, \dots, N_{\text{bet}}$ (trials on which the participant bet). Write $p(\cdot)$, $q(\cdot)$ for the participant and sequence of a given observation.

$$\text{Bet}_n \sim \text{Bernoulli}(\text{logit}^{-1}(\eta_n^B)), \quad (5)$$

$$\eta_n^B = \beta_0^B + u_{1,p(n)} + v_{q(n)}^B + (\beta_{\text{run}}^B + u_{2,p(n)}) \text{RunDum}_n + (\beta_{\text{imb}}^B + u_{3,p(n)}) \text{ImbMag}_n, \quad (6)$$

$$\text{Side}_m \sim \text{Bernoulli}(\text{logit}^{-1}(\eta_m^S)), \quad (7)$$

$$\eta_m^S = \beta_0^S + u_{4,p(m)} + v_{q(m)}^S + (\beta_{\text{run}}^S + u_{5,p(m)}) \text{RunCue}_m + (\beta_{\text{imb}}^S + u_{6,p(m)}) \text{ImbCue}_m. \quad (8)$$

Cues enter asymmetrically across equations. $\text{RunDum}_n \in \{0, 1\}$ indicates that the sequence ends in a run of at least two identical outcomes, and $\text{ImbMag}_n = |\#H - \#T|/6 \in [0, 1]$; both are unsigned measures of cue salience, since willingness to engage should depend on how conspicuous a sequence looks, not on which side it favors. In the side equation, $\text{RunCue}_m \in \{-1, 0, 1\}$ equals $+1$ (-1) if the sequence ends in a run of at least two heads (tails) and 0 otherwise, and $\text{ImbCue}_m = (\#H - \#T)/6 \in [-1, 1]$; both are signed toward heads, since direction is what side choice is about.

Participant-level random effects are drawn jointly:

$$\mathbf{u}_i = (u_{1,i}, \dots, u_{6,i})' \sim \mathcal{N}_6(\mathbf{0}, \Sigma), \quad \Sigma = \text{diag}(\tau) \Omega \text{diag}(\tau), \quad (9)$$

with indices 1–3 the (intercept, run-slope, imbalance-slope) of the betting equation and 4–6 the same triple for side choice. The cross-equation entries $\Omega_{1,4}$, $\Omega_{2,5}$, $\Omega_{3,6}$ constitute the dissociation test; the remaining off-diagonal entries capture within-equation dependence among a participant's own intercept and slopes. Independent per-sequence random

intercepts, uncorrelated with $\mathbf{u}_i$ and each other, are included in both equations:

$$v_s^B \sim \mathcal{N}(0, \sigma_{v^B}^2), \quad v_s^S \sim \mathcal{N}(0, \sigma_{v^S}^2). \quad (10)$$

Priors.

$$\beta_0^B, \beta_{\text{run}}^B, \beta_{\text{imb}}^B, \beta_0^S, \beta_{\text{run}}^S, \beta_{\text{imb}}^S \sim \mathcal{N}(0, 1.5^2), \quad (11)$$

$$\tau_k \sim \text{Student-}t_3^+(0, 1), \quad k = 1, \dots, 6, \quad (12)$$

$$\Omega \sim \text{LKJ}(2), \quad (13)$$

$$\sigma_{v^B}, \sigma_{v^S} \sim \text{Student-}t_3^+(0, 1). \quad (14)$$

Priors are weakly informative on the logit scale; a normal(0, 1.5²) prior on log-odds coefficients places negligible mass outside plausible behavioral effect sizes while leaving the estimates driven by the data.

Estimation. The model was implemented in Stan and fit via Hamiltonian Monte Carlo using `rstan`, with a non-centered parameterization for the participant-level random effects:

$$u_{k,i} = \tau_k \sum_{j \leq k} L_{kj} z_{j,i}, \quad LL' = \Omega, \quad z_{j,i} \sim \mathcal{N}(0, 1), \quad (15)$$

where L is the Cholesky factor of Ω , sampled directly via an LKJ Cholesky prior rather than sampling Ω itself. Sequence-level intercepts were similarly non-centered. Each treatment (FN, FP) was fit separately. Production runs used 4 chains of 8,000 iterations each (1,500 warmup, 6,500 post-warmup draws per chain, 26,000 draws total), with `adapt_delta` = 0.97 and `max_treedepth` = 13 to accommodate the funnel geometry typical of hierarchical variance parameters.

Convergence. Convergence was assessed via $\hat{R}$, effective sample size, and divergent transitions for all fixed effects, variance components, and correlation parameters, excluding the non-centered nuisance parameters z , u , and L , whose raw diagnostics are uninformative

under this parameterization. We require $\hat{R} \leq 1.01$ and an effective sample size of at least 400. No divergent transitions occurred in either treatment, and in FP every monitored parameter meets both criteria. In FN, all fixed effects, all τ_k , and all correlations used in the paper meet both criteria; the only exceptions are two correlations that enter no reported result.²

Posterior predictive checks. Rather than generating full per-trial replicated outcomes within Stan — infeasible at this draw count given the size of the trial-level data — posterior predictive checks were computed in R: 1,000 posterior draws of all fixed and random effects were subsampled, used to compute η^B and η^S for every observed trial, and simulated Bernoulli outcomes were compared against the observed aggregate betting and side-choice rates. The observed rates lie inside the 95% posterior predictive intervals in both treatments: betting rate 0.769 [0.761, 0.778] and share of heads choices 0.505 [0.492, 0.519] in FN; 0.682 [0.667, 0.698] and 0.524 [0.500, 0.547] in FP.

C.4.1 Results

	Bet = 1	ρ (Bet $\leftrightarrow$ Heads)	Heads = 1	
Has Terminal Run	0.26*	-.27	-0.79*	Terminal Run
{= 1}	(+2.7pp*)	[-.71, .28]	(-12.2pp*)	{-1 = TT, 0, 1 = HH}
Imbalance Magnitude	2.19*	-.17	0.34	Imbalance
{0, $\frac{1}{3}$, $\frac{2}{3}$, 1}	(+24.8pp*)	[-.46, .14]	(+7.8pp*)	{-1 (T) to 1 (H), by $\frac{1}{3}$ }
Intercept	0.69	.45	0.03	Intercept
	(57.5%)	[.02, .76]	(50.9%)	

Table C2: **Population-average fixed effects and cross-equation dissociation test, FP Treatment.** Same conventions as Table ???. Note the Imbalance Magnitude row, Heads = 1 column: the log-odds coefficient (0.34) is *not* credible, while the AME beside it (+7.8pp) *is* – not an error; this is the regressor with by far the largest participant-level heterogeneity relative to its population average (see main text).

²These are the posterior correlations of the participant-level betting run slope with the betting imbalance slope and with the side imbalance slope, with $\hat{R}$ of 1.012 and 1.036 and effective sample sizes of 240 and 139, respectively. They are weakly identified because participants barely differ in their betting run slope: its participant-level standard deviation is 0.30 (95% CrI [0.04, 0.52]), compared with 5.10 for the betting imbalance slope and 4.76 for the side imbalance slope.

Regressor	FN Treatment ($m = 2.5, n = 149$)		FP Treatment ($m = 1.9, n = 53$)	
	τ [95% CI]	$\tau/ \hat{\beta} $	τ [95% CI]	$\tau/ \hat{\beta} $
(Bet) Has Terminal Run	0.30 [0.04, 0.52]	2.87	0.45 [0.08, 0.83]	1.72
(Bet) Imbalance Magnitude	5.10 [4.30, 6.11]	1.69	4.08 [3.12, 5.40]	1.86
(Heads) Terminal Run	1.00 [0.82, 1.21]	1.07	1.06 [0.75, 1.48]	1.34
(Heads) Imbalance	4.76 [4.08, 5.61]	6.40	4.68 [3.56, 6.31]	13.59

Panel B: Participants with a credible personal side-choice slope

Run (against / with / uncertain)	60 (40%) / 1 (1%) / 88 (59%)	21 (40%) / 1 (2%) / 31 (58%)
Imbalance (against / with / uncertain)	28 (19%) / 45 (30%) / 76 (51%)	10 (19%) / 15 (28%) / 28 (53%)

Table C3: **Participant-level heterogeneity in run- and imbalance-sensitivity, by treatment.** Panel A: posterior SD (τ) of each participant-level random slope and its ratio to the absolute population fixed effect (Tables ?? and ??). Panel B: each participant’s personal side-choice slope (population fixed effect plus their own random-effect deviation), classified as credibly positive, credibly negative, or indistinguishable from zero at the 95% level, separately by treatment. The corresponding betting-equation classification is omitted for both treatments: essentially all participants are “uncertain” on the run slope (100% in FN, 94.3% in FP), and the imbalance slope shows no credible “disengaged” group in either treatment (1–2 of 149 in FN, 1 of 53 in FP, consistent with chance) – direction is not the informative axis for the extensive margin the way it is for side choice.

Class	n	Decision time (s)	Stake	Risk pref. (r)	Optimism (z)	Welfare loss
<i>FN Treatment</i>						
Against	28	7.8 (2.6)	45.2 (22.8)	0.73 (0.46)	−0.26 (1.04)	0.096 (0.127)
With	45	7.5 (2.4)	43.9 (25.4)	0.97 (0.51)	−0.10 (0.87)	0.116 (0.147)
Uncertain	76	7.1 (3.0)	50.4 (26.7)	0.88 (0.64)	0.08 (1.09)	0.182 (0.255)
<i>FP Treatment</i>						
Against	10	6.9 (1.2)	40.6 (28.6)	0.73 (0.49)	−0.45 (1.23)	0.066 (0.066)
With	15	7.3 (2.0)	47.5 (22.7)	0.93 (0.46)	0.08 (0.86)	0.135 (0.181)
Uncertain	28	6.3 (3.0)	47.2 (26.9)	0.90 (0.66)	0.33 (0.80)	0.202 (0.281)

Table C4

Sequence type	n seq.	Decision time (s)	Stake	Welfare loss
<i>FN Treatment</i>				
Imbalanced, with run	24	7.2 (5.9)	50.9 (31.5)	0.156
Imbalanced, without run	20	7.4 (6.5)	48.4 (31.4)	0.137
Balanced, with run	8	7.1 (6.1)	49.1 (31.0)	0.139
Balanced, without run	12	7.5 (6.3)	48.3 (31.6)	0.146
<i>FP Treatment</i>				
Imbalanced, with run	24	6.4 (5.1)	50.7 (31.4)	0.181
Imbalanced, without run	20	6.9 (5.9)	48.8 (30.4)	0.152
Balanced, with run	8	6.7 (5.9)	46.2 (30.7)	0.133
Balanced, without run	12	6.9 (6.0)	46.7 (29.8)	0.136

Table C5: **Decision time, stake, and welfare loss by sequence type.** Decision time and stake are pooled trial means (SD in parentheses) across all participant $\times$ sequence trials in that category; welfare loss is sequence-level μ_s^c averaged across the sequences in each category. By treatment. Balanced = $|\text{imbalance}| = 0$; imbalanced = $|\text{imbalance}| > 0$.

Table C6: Decision time, stake, and welfare loss by sequence imbalance and side bias (FN treatment)

Sequence type	n seq.	Decision time (s)	Stake	Welfare loss
Imbalanced, no side bias	16	7.3 (6.5)	50.9 (32.1)	0.160
Imbalanced, side bias	28	7.3 (6.0)	49.1 (31.0)	0.140
Balanced, no side bias	8	7.9 (6.6)	48.1 (31.6)	0.142
Balanced, side bias	12	7.0 (6.0)	48.9 (31.3)	0.144

C.5 Debrief Classification

After the 64 betting rounds, participants answered an open-ended debriefing question asking whether they had followed any rule or strategy while choosing a side to bet on and, if so, to describe it in their own words. Of the 202 participants, 99 gave a codable free-text description (FN, $m = 2.5$: $n = 73$; FP, $m = 1.9$: $n = 26$). Each description was hand-coded by a research assistant into one of five categories: **Pro-trend**, **Anti-trend**, **Unclear**, **No-trend**, and **Misc**:

- **Pro-trend.** Participants who specifically said they go *with* the majority or terminal symbol—i.e., that they bet on the pattern continuing.
- **Anti-trend.** Participants who specifically said they go *against* the majority or terminal symbol.
- **Unclear.** Participants who indicated *some* pro- or anti-trend tendency but did not specify which direction, together with participants who described following “intuition”—i.e., they followed some pattern or heuristic but did not say what it was.
- **No-trend.** Participants who indicated they knew the outcome was random (recognized statistical independence) and similar reasoning unrelated to any perceived pattern.
- **Misc.** Participants who did not talk about side choice at all, or who described something completely different from the betting task.

Counts (of $n = 99$ codable responses): Pro-trend 5, Anti-trend 17, Unclear 40, No-trend 12, Misc 25. The lists below give every codable response verbatim (original Czech, as recorded) with an English translation, identified by participant code (PID) and treatment (FN, $m = 2.5$; FP, $m = 1.9$), grouped by category.

Pro-trend ($n = 5$)

0awy2ukg (FN): *většinou jsem, i přes vědomí, že šance je 50/50 volila stranu která byla předtím vícekrát hozena* — Mostly, despite knowing the odds were 50/50, I chose the side that had come up more often before.

291cijhu (FN): *Když bylo více políček s “hlava” spíše jsem inklinovala k této odpovědi* — When there were more “heads” squares, I tended to lean toward that answer.

b6f2y62u (FN): *hot train - pokud poslední 2-3 hody byly O tak jsem inklinovaný k tomu věřit, že i další hod bude O. Pokud poslední hody neodpovídaly tomuto vzorci, snažil jsem se “vyvážit” všechny hody do poměru 50:50 — “Hot train”—if the last 2–3 tosses were tails, I was inclined to believe the next toss would also be tails. If the recent tosses did not fit this pattern, I tried to “balance” all the tosses to a 50:50 ratio.*

wvles9rw (FN): *vždy jsem sázela na to, co se nacházelo na posledním políčku. — I always bet on whatever was in the last square.*

zhwrqfgq* (FN): *Pokud se sekvence zdála, že bude pokračovat, tak byla větší pravděpodobnost, že vsadím — If the sequence seemed like it would continue, I was more likely to bet.*

Anti-trend ($n = 17$)

1ai0bv74 (FN): *Pokud padalo hodně H, věřila jsem, že situace se obrátí v O. A naopak. Pokud historie jevila periodicitu, snažila jsem se ji porušit a utvrdit na střídačku mezi koly. (50/50) — If a lot of H came up, I believed the situation would turn to O, and vice versa. If the history showed periodicity, I tried to break it and stick to alternating between rounds (50/50).*

5zmzfp5u (FN): *Když se jedno zobrazilo méně než to druhé, zvolil jsem ho. — When one outcome showed up less than the other, I chose it.*

8u9yzqpm (FP): *jestli se v sekvenci objeví H nebo O vícekrát jak 3 zvolím opak — If H or O appears more than 3 times in the sequence, I choose the opposite.*

9jzw9zw9 (FN): *Pokud bylo příliš hodů jedné strany, vybíral jsem možnost druhou. — If there were too many tosses of one side, I picked the other option.*

ab9845tr (FN): *většinou jsem sázel na to čeho bylo v historii hodů méně — I mostly bet on whichever there was less of in the toss history.*

cosykh2d (FN): *Předpokládala jsem pravděpodobnost hodu obou stran na 50%. Když bylo více H, zvolila jsem O a naopak. — I assumed the probability of each side was 50%. When there were more H, I chose O, and vice versa.*

cq79aqdk (FN): *Toho čeho bylo méně při “hodu mincí” když jsou šance 50/50 by měla být celkově rovnoměrná distribuce. 50% hranice úspěšnosti v druhé části — Whichever there was less of during the “coin toss,” when the odds are 50/50, the overall distribution should be even. 50% success threshold in the second part.*

d6atlcg6 (FN): *vždy jsem volila tu minci, která se v sekvenci objevovala méně často a podle toho, o kolik, jsem určovala, kolik vsadím; pokud se H i O objevily stejně často, většinou jsem nesázela — I always chose the coin side that appeared less often in the sequence, and determined how much to bet based on the size of that gap; if H and O appeared equally often, I mostly did not bet.*

gmcai22c (FN): *Příznám se, že jsem se nechal/a ovlivnit sekvencí - ačkoliv vím, že na pravděpodobnost*

dalšího hodu nemají vliv, pokud bylo více hlav tak jsem zvolil/a orla a naopak. Pokud bylo víceméně stejně hlav a orlů, tak jsem nesázal/a. — I admit I let myself be influenced by the sequence—even though I know it has no effect on the probability of the next toss—if there were more heads I chose tails, and vice versa. If heads and tails were roughly equal, I did not bet.

iw5t8dyo (FN): *vzhledem ke zobrazené sekvenci jsem šla “proti” nejčastěji zobrazovanému symbolu, protože hodit stejný symbol 7x za sebou nebo házet tak, aby v 5 ze 7 hodů padl jeden symbol není zas tak pravděpodobné i když je mince spravedlivá. Jinak jsem udržovala celkem nízké sázky, protože je lepší málo vyhrát než hodně prohrát* — Given the displayed sequence, I went “against” the most frequently shown symbol, because getting the same symbol 7 times in a row, or having one symbol come up in 5 of 7 tosses, is not all that likely even for a fair coin. Otherwise I kept my stakes fairly low, because it is better to win a little than lose a lot.

lwdhj1kl (FN): *I když logicky vím, že mince je spravedlivá, aplikoval jsem pravidlo na bázi předchozích hodů. Pokud byl počet H a O stejný, nesázím. Pokud byl 4 ku 2, sázím 50 ECU na méně zastoupenou minci. Pokud byl 5 ku 1, sázím 75 ECU. Pokud byl 6 ku 0, sázím 100 ECU.* — Even though I logically know the coin is fair, I applied a rule based on the previous tosses. If the count of H and O was equal, I do not bet. If it was 4 to 2, I bet 50 ECU on the less-represented side. If it was 5 to 1, I bet 75. If it was 6 to 0, I bet 100.

p4alad9n (FN): *Vycházel jsem z počtu hozených stran mincí v uvedené řadě. Pokud byly obě strany uvedeny v stejné míře (3H a 3O), pak jsem nesázal. Pokud byla jedna strana zastoupena méně, pak jsem na ni vsadil. Čím méně byla jedna strana zastoupena, tím více jsem na ní vsadil, protože byla větší pravděpodobnost, že v dalším hodu padne.* — I went by the count of sides thrown in the given sequence. If both sides were represented equally (3H and 3O), I did not bet. If one side was less represented, I bet on it. The less a side was represented, the more I bet on it, because there was a greater probability it would come up on the next toss.

plvbe2gd (FP): *To, co padlo výrazně méněkrát, jsem volila.* — I chose whatever had come up markedly less often.

pnssdock (FN): *Vsázal jsem vyšší částky na symbol který se v sekvenci objevoval méně, ne nutně kvůli efektivitě, spíš pro zábavu.* — I bet larger amounts on the symbol that appeared less in the sequence—not necessarily for efficiency, more for fun.

sqtwsk3a (FN): *když se hlava/orl opakovala zvolila jsem opak* — When heads/tails repeated, I chose the opposite.

wglyejt5 (FN): *zvolím takovou stranu mince která je uvedena méněkrát, i když vím že to šanci nijak neovlivní.* — I choose whichever side of the coin is listed fewer times, even though I know it does not affect the odds at all.

wlrphv6e (FN): *hod mincí by se měl co nejvíce přiblížit 50% na obě možnost, tedy pokud je nějaká*

možnost jasně zastoupena více, je logičtější zvolit možnost druhou, zároveň jsem hleděl na pravděpodobnost opakování stejných hodů, tedy pokud byla jedna možnost zastoupena několikrát za sebou jako poslední hody, měla by být sekvence narušena — A coin toss should come as close as possible to 50% for both options, so if one option is clearly more represented, it is more logical to choose the other one; I also considered the probability of the same outcome repeating, so if one option had come up several times in a row in the most recent tosses, the sequence ought to be “disrupted.”

Unclear ($n = 40$)

2n8v21wa (FN): *snažila jsem se najít nějakou spojitost a řád a přemýšlela nad pravděpodobností* — I tried to find some connection and order, and thought about the probability.

45tler5z (FN): *Vybírala jsem instinktivně, podle toho, jak se tam zrovna dané písmeno hodilo podle předchozích.* — I chose instinctively, based on how the given letter happened to fit with the previous ones.

4i218wdr (FN): *Moje strategie byla čistě na pocitu.* — My strategy was purely based on feeling.

5nhcscxw (FN): *Jednoduše co mi v hlavě znělo lépe. Co mělo lepší rytmus. Trochu absurdní přístup, ale dával mi smysl.* — Simply whatever sounded better in my head. Whatever had a better rhythm. A bit of an absurd approach, but it made sense to me.

5nib79gb (FN): *Opakující se vzorec v hodech* — A repeating pattern in the tosses.

63fxhvj1 (FN): *čistě podle sekvence, doufal jsem, že to bude nějak navazovat, i když to bylo 50/50* — Purely based on the sequence—I hoped it would somehow continue, even though it was 50/50.

61zqnj5b (FP): *Čistě podle padlých předchozích hodů. Podle toho, jaký jsem viděl poměr mezi mincemi.* — Purely based on the previous tosses that came up. Based on the ratio I saw between the two coins.

7nit27ae (FN): *Přesto že logicky historie absolutně neovlivňuje to, co padne, občas přihlížím na délku toho, kolik z posledních hodů mělo stejnou stranu. Podle toho jsem sázel.* — Even though logically the history has absolutely no effect on what comes up, I sometimes look at how many of the recent tosses in a row had the same side. I based my bet on that.

8aac14nd (FN): *Občas slyším hlasy v hlavě, nebo mi přijde, že mám vidiny toho, co by tam mohlo být. Řídil/a jsem se tím.* — Sometimes I hear voices in my head, or feel like I am having visions of what might come up. I went by that.

azb2vhuy (FN): *Jít za jednou věcí, která mi připadala jistá* — Go with the one thing that seemed certain to me.

br30vc7y (FN): *Logické usuzování, dle pořadí „historie“ hodů.* — Logical reasoning, based on the order of the tosses’ “history.”

bw11i1zf (FN): *Dle pocitu* — By feeling.

cog7g3d5 (FN): *Snažila jsem se hledat logiku* — I tried to look for logic.

dbf17k03 (FP): *Patern který jsem dokázal vidět, ikdyž tam nebyl.* — A pattern I could make out, even though it was not really there.

dj9y5dzh (FN): *Jestli 50-50 hlavy a orly, tak nebrala jsem nic* — If it was 50-50 heads and tails, I did not take anything [did not bet].

dn2att26 (FN): *Pokud jsem viděla v “hodech” pravidlo, dala jsem větší sázku na hod, který by korespondoval s “pravidlem”.* — If I saw a rule in the “tosses,” I placed a larger stake on the outcome that matched that “rule.”

dzek6s7t (FN): *Hledal jsem v hodech jistý pattern protože jsem předpokládal že se bude opakovat a na to jsem pak vsadil částku kterou bych byl ochoten ztratit za předpokladu reálných peněz.* — I looked for a certain pattern in the tosses because I assumed it would repeat, and I bet an amount I would have been willing to lose, assuming it were real money.

e9nv0559 (FP): *pokud jsem viděla v zobrazených hodech nějakou pravidelnost, tak jsem v ní pokračovala* — If I saw some regularity in the displayed tosses, I continued it.

fae6a43n (FP): *Je-li šance 50% nesázím, jsou-li 4 O dávám málo, je-li 5 O dávám 20 % je-li 6 O dávám 50- 100%* — If the chance is 50%, I do not bet; if there are 4 tails, I put in a little; if there are 5 tails, I put in 20%; if there are 6 tails, I put in 50–100%.

hv2emsnb (FN): *ze začátku jsem se snažil předvídat dle historie, ke konci jsem se rozhodl držet spíše maximální sázky na hlavu a pouze v případech kdy mi přišlo nepravděpodobné že by padla tak jsem sázku změnil* — At the start I tried to predict based on history; toward the end I decided to mostly hold maximum bets on heads, and only changed my bet in cases where it seemed unlikely that heads would come up.

hvzz621s (FN): *podívala jsem se na předešlé výsledky a řekla jsem si, jak velká je třeba pravděpodobnost, že to padne znovu.* — I looked at the previous results and told myself roughly how high the probability was that it would come up again.

i023mf33 (FP): *navázal jsem na sekvenci, jako by se opakovala* — I continued the sequence, as if it would repeat.

im31dsss (FN): *vždycky jsem volila, podle toho co mi jako první vyběhlo v hlavě když jsem dořekla poslední kostku, když tam nebylo nic nesázela jsem* — I always chose based on whatever first popped into my head right after I finished reading the last square; if nothing came to mind, I did not bet.

jgehwy25 (FP): *vzor předešlých hodů* — The pattern of the previous tosses.

ly9fbrhn (FP): *Pokud z historie hodů vypadala, že by bylo pravděpodobnější, aby padla daná strana, tak jsem sázela. Pokud ne, tak jsem nesázela. Vždy jsem si ale ponechala cca polovinu měny u sebe.* — If the toss history made it look like a given side was more likely to come up, I bet [on it]. If not, I did not bet. But I always kept about half the currency for myself.

o6jjy2a8 (FN): *Přestože vím, že je mince spravedlivá, rozhodovala jsem podle vzorů, co jsem v historii*

našla, abych nemusela náhodně rozhodovat mezi možnostmi. — Even though I know the coin is fair, I decided based on patterns I found in the history, so that I would not have to choose between the options at random.

ohftsoi (FP): *Mince sice byla spravedlivá, ale odměna nebyla dvojnásobná. Vzhledem k pravděpodobnosti 50% při každém vsázce jsem tedy statisticky přicházel o peníze. Vsadil jsem si až teprve pokud jsem viděl delší sekvenci stejných hodů za sebou (3 a více) s plným vědomím, že se jedná o gamble* — The coin was indeed fair, but the reward was not doubled. Given the 50% probability on every bet, I was therefore statistically losing money. So I only placed a bet once I saw a longer sequence of the same outcome in a row (3 or more), fully aware that it amounted to a gamble.

rsbmid96 (FN): *intuice* — Intuition.

scvzygig (FN): *Pokud jsem v předchozích hodech viděla nějaký opakování a byla jsem si tím jista sázela jsem více.* — If I saw some repetition in the previous tosses and was confident about it, I bet more.

skhgdzt1 (FN): *Nesázet nikdy vše, podle řady podle počtu hlav nebo orlů určovat výšší sázky, 2 z 6 kolem 40, 1 z 6 kolem 50 a 0 z 6 kolem 60* — Never bet everything; determine the stake size from the count of heads or tails in the sequence: 2 of 6, around 40; 1 of 6, around 50; 0 of 6, around 60.

tbcjq3pk (FN): *podle pocitu* — By feeling.

vf1rugtz (FN): *snažil jsem se v padlých stranách najít nějakou sekvenci* — I tried to find some sequence in the outcomes.

vmyv36qs (FP): *nejsem kdejaký stratég, ale považovala jsem za dobré zohlednit pravděpodobnosti v historii a pozorovat, jak se projevuje náhoda, např. pokud byly hody HOOHOO –ž dala jsem vysokou výši odměny na Hlavu, pokud byla historie iracionální (OHHHHH), poněkud mě to v daném kole odradilo od vysoké nebo jakékoli sázky* — I am no great strategist, but I thought it good to take the probabilities in the history into account and observe how randomness manifests—e.g., if the tosses were H-O-O-H-O-O, I placed a high stake on Heads; if the history looked “irrational” (O-H-H-H-H-H), that somewhat discouraged me from placing a high stake, or any stake, that round.

x0udfh66 (FN): *Snažila jsem se zohledňovat obě strany mince* — I tried to take both sides of the coin into account.

x6dcmxpj (FN): *doplňování vzorce nebo podle pocitu* — Completing the pattern, or going by feeling.

xddoq4wr (FN): *mám ráda když jsou věci pravidelné a do počtu, když jsem vyčetla z předchozích hodů nějakou pravidelnost nebo vzor - vsadila jsem tak, aby vzor pokračoval nebo se opakoval* — I like it when things are regular and add up; when I read some regularity or pattern out of the previous tosses, I bet so that the pattern would continue or repeat.

xhohlmv4 (FN): *Vybírala jsem pocitově možnost, ke které mě nejvíc táhla intuice.* — I chose by feeling, whichever option my intuition pulled me toward most.

y7hwht9z (FP): *Snaha nalézt posloupnost* — An attempt to find a sequence.

ylphslum (FN): *většinou podle počtu padlých stran* — Mostly based on the count of each side that had come up.

z4gf3qxy (FN): *když byla historie hodů 3O a 3H, spíš jsem nesázela, když to bylo 2/4 tak jsem sázela 10 a když 1/5 nebo 0/6 tak jsem sázela 25* — When the toss history was 3O and 3H, I mostly did not bet; when it was 2/4 I bet 10; and when it was 1/5 or 0/6 I bet 25.

No-trend ($n = 12$)

0me0e9oy (FN): *Nesázel jsem a tím jsem měl jistotu 100%, že dostanu původní částku* — I did not bet, and that way I had 100% certainty of getting the original amount.

4pztp3n7 (FN): *v posledních dvou jsem vsázel buď čistě na “hlavu” a nebo ve druhém pouze na “orla”* — In the last two [blocks], I bet either purely on “heads,” or in the other only on “tails.”

6a5rwuf0 (FP): *Statisticky jsem si určil jestli se vyplatí sázet. U hodu mincí se to nevyplatí u procent jsem odhadl že od 45 už se vyplatí jít do B.* — I worked out statistically whether it was worth betting. For a coin toss it is not worth it; for percentages, I estimated that from 45 upward it becomes worth going for B.

h2il1g8k (FN): *50% šance na oba, takže jsem se rozhodl v každé části dát 100% na buďto vždy hlavu nebo orla* — 50% chance for both, so I decided in each block to put 100% on always heads or always tails.

io5cm2g0 (FN): *hlava nebo nesázet* — Heads, or do not bet.

j8hqk15c (FN): *Každý blok jsem se snažila vsadit na hlavu i orla a také si nevsadit.* — In every block I tried to bet on both heads and tails, and also to not bet.

q27arz13 (FN): *Vždy jsem sázel při hodu mincí na stranu orla a nikdy jsem nesázel všechen rozpočet. Částky jsem měnil na základě osobního “pocitu”, ale jinak jsem své chování neměnil.* — I always bet on tails when tossing the coin, and never bet my entire budget. I varied the amounts based on personal “feeling,” but otherwise did not change my behavior.

rhwxumg (FP): *I přestože je dobré koukat na skupinu hodů jako na celek a na základě toho se rozhodovat před učiněním hodů, tak předešlé hody už proběhly, tudíž pravděpodobnost H/O se na finálovém hodu neprojeví a šance je doopravdy 50/50 a vzhledem k tomu, že se sázka násobí 1.9x, je naprosto nevýhodné statisticky vsázet.* — Even though it is good to look at a group of tosses as a whole and decide on that basis before the tosses are made, the earlier tosses have already happened, so H/O probability is not reflected in the final toss and the chance really is 50/50; and given that the stake is multiplied by 1.9x, it is statistically completely disadvantageous to bet.

u93581v4 (FN): *Prislo m logicky nikdy nesazet. Ale nekolikrat jsem sazel pro ‘zabavu’ kdybych dostal odmenu za vsechny kola, vsude bych vybral H* — It seemed logical to me to never bet. But I bet several

times “for fun”—if I had gotten a reward for every round, I would have picked H every time.

v64rioaz (FP): *sazet hodně na jednu stranu, riskovat* — Bet a lot on one side, take risks.

x4hw31zf (FN): *nejdříve jsem se rozhodovala na základě předchozích hodů. Poté už jen na základě 50% pravděpodobnosti. Představila jsem si opravdový hod mincí a došlo mi, že pokud simulace skutečně odráží tuto realitu, nic na tom asi nevykoumám :-)* — At first I decided based on the previous tosses. Then only based on the 50% probability. I pictured an actual coin toss and realized that if the simulation really reflects that reality, I probably will not figure anything out about it :-)

xm30b9oy (FP): *Zvolím jeden znak a ten “držím” čímž si alespoň zajistím to, že si budu pamatovat na co jsem jakou částku vsadila. Jestli Vám to nějak pomůže.* — I pick one symbol and “hold” it, which at least ensures I remember which amount I bet on what. If that helps you somehow.

Misc ($n = 25$)

0sju5bel (FN): *když nevíš, dávej za B* — When you do not know, put it on B.

1bye37zt (FN): *nesázet moc, pokud tomu nevěřím* — Do not bet much if I do not believe in it.

1zntsgei (FN): *strategie byla jednoznačná mám 64 pokusů kde mince má být spravedlivá a mám “neomezené” finanční prostředky tím pádem 1:128 že dostanu plnou výhru neb nulu ze začátku jsem to hrál spíše na jistotu ale pak utvrdil strategii a vždy sázel 100% sázky na volbu dle pocitu / náhody.* — The strategy was straightforward: I have 64 attempts where the coin is supposed to be fair and I have “unlimited” funds, so there is a 1:128 chance of getting the full win or zero; at the beginning I played it more for certainty, but then I settled on a strategy and always bet 100% of the stake on a choice made by feeling / at random.

291gvzas (FN): *myslel jsem, že obě experimentální části budou totožné. Chtěl jsem v první riskovat a v druhé naopak vůbec nesázet. To by znamenalo, že bych měl šanci něco získat + mít vyšší odměnu již jistou* — I thought both experimental parts would be identical. I wanted to take risks in the first one and, on the contrary, not bet at all in the second. That would mean I would have a chance of gaining something plus already having a higher reward guaranteed.

3aizilzu (FN): *Jít do co nejmenšího rizika* — Go for as little risk as possible.

4cfcyav8 (FN): *alespoň v polovině ze sekvencí nesázet a když už vsadit tak všechno.* — Not bet in at least half of the sequences, and if I did bet, bet everything.

6jr1sfdd (FP): *pokud to nebylo 10/90 tak jsem nesázela až na pár případů* — If it was not [as skewed as] 10/90, I mostly did not bet, except in a few cases.

7m0ipsjl (FN): *opakování písmen* — Repetition of letters.

ax1hs4wd (FP): *lepší jít na jistotu, než prohrát vše.* — Better to go for certainty than lose everything.

cilzf61z (FN): *že každý krát byl stejná 50/50* — That every time it was the same 50/50.

d9qtaz7t (FP): *Jelikož to byli imaginární peníze, nebrala jsem to tak vážně a rádeji riskovala.* —

Since it was imaginary money, I did not take it too seriously and preferred to take risks.

glksvai9 (FP): *nejde o peníze ale o studii.tudiz jsem zkousela studovat vase metody vic nez to,jak se rozhodnu odpovedet. bylo napsano ze nase odmena bude 150 kc za ucast takze jakakoliv hra uz neni hrou nebot vysledek je predem urcen.* — It is not about the money, it is about the study, so I tried to study your methods more than how I would decide to answer. It was written that our reward would be 150 CZK for participation, so any game is no longer really a game, since the outcome is predetermined anyway.

iqcw2551 (FN): *skoro vždy mi něco zbyde, takže v každém případě s vydělám i variabilní složku* — Almost every time I have something left over, so either way I also earn the variable component.

j2uk4906 (FP): *Každý druhý hod jsem neriskovala a nevsadila.* — Every second toss I did not take the risk and did not bet.

jpvt91dp (FN): *pokud je riziko 50% a větší, je lepší vsadit na jistotu* — If the risk is 50% or more, it is better to bet on certainty.

kjkw3cqg (FN): *kdo si nevsadí, nevyhraje :D* — Nothing ventured, nothing gained :D (lit. “who does not bet, does not win”)

nmy9850u (FN): *Lepší vrabec v hrsti, než holub na střeše* — A bird in the hand is worth two in the bush (Czech proverb, lit. “better a sparrow in the hand than a pigeon on the roof”).

p781v0gk (FN): *Pokud pravděpodobnost výhry je 50 procent (podle mého názoru je velká) a nic neztratím, je lepší buď nikdy nevsazovat a mít jistých 100, a nebo s 50 procentní pravděpodobností zkusit získat 250. Ostatní strategie posle mě nejsou relevantní, a pokud neztrácím vlastní peníze tak mě přišlo že 50 procentní možnost výhry je lepší* — If the probability of winning is 50 percent (which in my opinion is high) and I lose nothing, it is better to either never bet and have a sure 100, or try with a 50 percent probability to get 250. Other strategies did not seem relevant to me, and since I was not losing my own money, a 50 percent chance of winning seemed better to me.

q5tvbdfb (FP): *Hrát na jistotu. 100 EUC je vždycky lepší než méně. Raději jsem počítal s fixní odměnou 100 EUC než bych riskoval s takto extrémně malou pravděpodobností, že ji ztratím.* — Play it safe. 100 ECU is always better than less. I preferred to count on a fixed reward of 100 ECU rather than risk it with such an extremely small probability of losing it.

sah158u2 (FP): *Pozůstatky znalostí kombinatoriky z gymnázia.* — Leftover knowledge of combinatorics from high school.

tb51st8n (FP): *U každé úlohy jsem zvolil hodnotu 50 s tím, že se buď usměje štěstí, nebo ne.* — For every task I chose the value 50, on the basis that luck would either smile on me or not.

ve31gbhb (FP): *SMART, myšlenkové mapy, tabulka plusů a mínusů Hod mincí, kdy nám v letu dojde, co vlastně chceme doopravy* — SMART, mind maps, a pros-and-cons table. A coin toss where, in the moment, we realize what we actually want.

vee8v72t (FP): Šel jsem proti statistice a rozhodl jsem se ne vždy se rozhodovat podle posledních 6 hodů. Tzv. jít proti trendu — I went against statistics and decided not to always decide based on the last 6 tosses—so-called going “against the trend.”

zdb1drk1 (FN): Neměla jsem co ztratit, proto mi v tuto chvíli nevadilo podstupovat finanční riziko. — I had nothing to lose, so at that point I did not mind taking on financial risk.

zjpkqjxq (FN): Má strategie je buď a nebo. Výsledek je náhoda, to neovlivním. A protože ty peníze nejsou fyzicky moje, tak mohu riskovat. A když vyhraju jen 1 Kč, furt mám o 1 Kč víc, než když jsem přišla. — My strategy is either/or. The outcome is random, I cannot influence it. And since the money is not physically mine, I can take risks. And if I win even 1 CZK, I still have 1 CZK more than when I arrived.

Table C7: Self-reported strategy use by model-derived slope classification

<i>Panel A: Imbalance-slope class</i> ($\chi^2_{(4)} = 7.44, p = .114$)				
Self-reported strategy use	Imbalance-slope class			Total
	With-imbalance	Uncertain	Against-imbalance	
No	24 (38.1%)	31 (49.2%)	8 (12.7%)	63
Unsure	9 (30.0%)	18 (60.0%)	3 (10.0%)	30
Yes	27 (24.8%)	55 (50.5%)	27 (24.8%)	109
Total	60 (29.7%)	104 (51.5%)	38 (18.8%)	202
<i>Panel B: Run-slope class</i> ($\chi^2_{(4)} = 5.83, p = .212$)				
Self-reported strategy use	Run-slope class			Total
	With-run	Uncertain	Against-run	
No	1 (1.6%)	38 (60.3%)	24 (38.1%)	63
Unsure	0 (0.0%)	23 (76.7%)	7 (23.3%)	30
Yes	1 (0.9%)	58 (53.2%)	50 (45.9%)	109
Total	2 (1.0%)	119 (58.9%)	81 (40.1%)	202

Notes: Cell entries are counts, with row percentages in parentheses; percentages may not sum to 100 due to rounding. $N = 202$ in both panels. Classification in each panel is derived from each participant’s fitted model slope on the relevant cue (imbalance-slope in Panel A, run-slope in Panel B). The reported test in each panel is a 3×3 Pearson χ^2 test of independence on the full sub-table. Neither panel’s test supports an association between self-reported strategy use and model-derived slope classification. With-run is a near-null class in the full sample ($\approx 1\%$), so the sparse with-run column in Panel B (0–1 observations per row) contributes little to that test there.

Table C8: Self-reported strategy category versus model-derived run-slope class (pooled across treatments, $n = 97$). Almost no participant's fitted run slope falls in the *with_run* direction (1 of 97), so no directional Fisher tests are reported; the table is descriptive only.

Category	With Run	Uncertain	Against Run	N
Pro-trend	1	2	1	4
Anti-trend	0	8	9	17
No-trend	0	10	2	12
Misc	0	17	8	25
Unclear	0	15	24	39
Total	1	52	44	97

C.6 Synthesis — Regressions

C.6.1 Model Specifications

Table C9: Determinants of Betting, Side Choice, and Stake (m25)

	<i>Dependent variable:</i>		
	Bet = 1	Stake (ECU)	Heads = 1
	(1)	(2)	(3)
Imbalance Magnitude [0 (balanced) to 1 (pure)]	1.970*** (0.127)	9.578*** (0.899)	
Terminal Run of 2 Present [binary]	0.012 (0.061)	1.049** (0.465)	
Imbalance [−1 (T-Pure) to +1 (H-Pure)]			0.471*** (0.072)
Terminal Run [−1 = TT, +1 = HH, 0 = No Run]			−0.654*** (0.043)
Constant	1.657*** (0.225)	43.642*** (2.135)	0.034 (0.048)
Model	Logistic	Linear	Logistic
Participants (pid)	149	147	147
ICC (participant)	0.667	0.634	0.068
AIC	7200.2	64836.5	9763.3
BIC	7228.9	64871	9790.9
Observations	9,536	7,333	7,333

*p<0.1; **p<0.05; ***p<0.01

C.7 ...

C.7.1 Collinearity Diagnostics

Table C17 reports variance inflation factors (VIF) for the two fixed-effect predictors — imbalance and the terminal run indicator — across all three specifications and both treatments. Collinearity is low throughout: VIF is 1.04 in both treatments for the extensive-margin (betting) model, 1.06 (m25) and 1.07 (m19) for the intensive-margin (stake) model, and 1.57 (m25) and 1.61 (m19) for the side-choice model — all comfortably below the

Table C10: Determinants of Betting, Side Choice, and Stake (m19)

	<i>Dependent variable:</i>		
	Bet = 1 (1)	Stake (ECU) (2)	Heads = 1 (3)
Imbalance Magnitude [0 (balanced) to 1 (pure)]	2.008*** (0.205)	13.890*** (1.433)	
Terminal Run of 2 Present [binary]	0.193* (0.100)	0.137 (0.751)	
Imbalance [−1 (T-Pure) to +1 (H-Pure)]			0.367*** (0.126)
Terminal Run [−1 = TT, +1 = HH, 0 = No Run]			−0.595*** (0.077)
Constant	0.620 (0.386)	41.103*** (3.681)	0.087 (0.064)
Model	Logistic	Linear	Logistic
Participants (pid)	53	50	50
ICC (participant)	0.688	0.684	0.032
AIC	2749.6	19982.3	3120
BIC	2774.2	20011	3143
Observations	3,392	2,312	2,312

*p<0.1; **p<0.05; ***p<0.01

Table C11: Participant-level cue responsiveness and welfare loss

	Dependent variable: $\log \hat{\mu}_i^c$			
	(1)	(2)	(3)	(4)
Run responsiveness	-0.254 [-0.819, 0.155]	-0.312 [-0.704, 0.031]	-0.297 [-0.716, 0.055]	-0.299 [-0.730, 0.069]
Imbalance responsiveness	-0.066 [-0.213, 0.058]	-0.034 [-0.146, 0.063]	-0.030 [-0.147, 0.067]	-0.031 [-0.145, 0.072]
Number of bets		0.050* [0.025, 0.082]	0.050* [0.024, 0.082]	0.050 [-0.117, 0.227]
Side preference $ \hat{\mu}_i^h - \bar{h} $			3.291 [-7.793, 12.362]	3.244 [-7.641, 12.497]
Betting rate $\hat{\mu}_i^b$				0.217 [-11.090, 11.020]
Spearman ρ (run resp., loss)		-0.19	[-0.46, 0.12]	
Spearman ρ (imb. resp., loss)		-0.22	[-0.48, 0.06]	
Participants	50	50	50	50

Notes: Run responsiveness is minus the participant's personal side-choice slope on the signed terminal-run cue; imbalance responsiveness is the absolute personal slope on signed imbalance (both from the joint synthesis model). Welfare loss $\hat{\mu}_i^c$ (share of the endowment), betting rate, and side preference come from the RQ3, RQ1, and RQ4 models. Each statistic is computed within each of 1000 posterior draws, combined with Bayesian-bootstrap weights over participants; cells report the posterior median and 95% credible interval, and an asterisk denotes an interval excluding zero. Because the inputs come from separately estimated models, correlations are attenuated toward zero. Full sample.

Table C12: Participant-level cue responsiveness and welfare loss

	Dependent variable: $\log \hat{\mu}_i^c$			
	(1)	(2)	(3)	(4)
Run responsiveness	-0.247 [-0.584, 0.035]	-0.243 [-0.580, 0.054]	-0.237 [-0.587, 0.056]	-0.236 [-0.594, 0.051]
Imbalance responsiveness	-0.068* [-0.142, -0.006]	-0.065* [-0.142, -0.004]	-0.061 [-0.137, 0.003]	-0.061 [-0.139, 0.003]
Number of bets		0.004 [-0.013, 0.019]	0.004 [-0.012, 0.019]	0.005 [-0.097, 0.113]
Side preference $ \hat{\mu}_i^h - \bar{h} $			0.990 [-2.472, 4.402]	0.960 [-2.578, 4.452]
Betting rate $\hat{\mu}_i^b$				-0.067 [-7.003, 6.540]
Spearman ρ (run resp., loss)		-0.19	[-0.40, 0.04]	
Spearman ρ (imb. resp., loss)		-0.22	[-0.41, 0.00]	
Participants	95	95	95	95

Notes: Run responsiveness is minus the participant's personal side-choice slope on the signed terminal-run cue; imbalance responsiveness is the absolute personal slope on signed imbalance (both from the joint synthesis model). Welfare loss $\hat{\mu}_i^c$ (share of the endowment), betting rate, and side preference come from the RQ3, RQ1, and RQ4 models. Each statistic is computed within each of 1000 posterior draws, combined with Bayesian-bootstrap weights over participants; cells report the posterior median and 95% credible interval, and an asterisk denotes an interval excluding zero. Because the inputs come from separately estimated models, correlations are attenuated toward zero. Confirmatory sample.

Table C13: Sequence features and outcomes across the four margins

	Direction	Engagement and welfare		
	Side bias	Betting rate	Stake deviation	Welfare loss
	$ \hat{\mu}_s^h - \bar{h} $	$\hat{\mu}_s^b$	$\hat{\mu}_s^a$	$\hat{\mu}_s^c$
Terminal run	1.8 [−0.8, 4.3]	1.6 [−0.3, 3.7]	0.2 [−0.9, 1.1]	0.1 [−0.8, 1.0]
Imbalance 2–4	−0.4 [−3.2, 2.4]	6.8* [4.3, 9.5]	0.2 [−1.0, 1.4]	1.0* [0.0, 2.0]
Imbalance 1–5	−1.9 [−5.2, 1.6]	10.3* [7.1, 13.6]	3.9* [2.2, 5.4]	5.3* [3.8, 6.9]
Imbalance 0–6	−3.3 [−9.1, 3.8]	14.1* [8.5, 20.0]	11.1* [8.0, 14.2]	24.8* [19.0, 32.9]
Constant (balanced, no run)	7.3* [4.4, 10.3]	61.8* [59.3, 64.1]	40.6* [39.4, 41.7]	13.4* [12.4, 14.5]
Sequences	64	64	64	64
Posterior draws	1000	1000	1000	1000

Notes: Each column regresses a sequence-level quantity on a terminal-run indicator (last two symbols identical) and imbalance-class indicators (reference: balanced 3–3), separately within each posterior draw. Cells report the posterior median of the coefficient (percentage points) with its 95% credible interval in brackets; an asterisk denotes an interval excluding zero. Side bias uses the full sample; betting rate, stake deviation, and welfare loss (shares of the endowment) use the full sample. The 0–6 class contains only the two pure sequences.

Table C14: Sequence features and outcomes across the four margins

	Direction	Engagement and welfare		
	Side bias	Betting rate	Stake deviation	Welfare loss
	$ \hat{\mu}_s^h - \bar{h} $	$\hat{\mu}_s^b$	$\hat{\mu}_s^a$	$\hat{\mu}_s^c$
Terminal run	5.6* [3.8, 7.6]	0.8 [−0.8, 2.3]	0.1 [−0.7, 0.9]	−0.4 [−1.0, 0.1]
Imbalance 2–4	−2.0* [−3.9, −0.1]	13.7* [11.9, 15.5]	−0.5 [−1.4, 0.5]	−1.7* [−2.4, −1.0]
Imbalance 1–5	−4.5* [−7.3, −1.8]	16.8* [14.4, 19.2]	3.1* [2.1, 4.4]	0.0 [−0.9, 0.9]
Imbalance 0–6	−9.2* [−13.3, −4.4]	21.4* [17.2, 25.2]	11.4* [9.0, 13.8]	15.5* [11.9, 19.9]
Constant (balanced, no run)	7.4* [5.6, 9.1]	63.6* [62.0, 65.2]	6.6* [5.7, 7.3]	13.0* [12.2, 14.2]
Sequences	64	64	64	64
Posterior draws	1000	1000	1000	1000

Notes: Each column regresses a sequence-level quantity on a terminal-run indicator (last two symbols identical) and imbalance-class indicators (reference: balanced 3–3), separately within each posterior draw. Cells report the posterior median of the coefficient (percentage points) with its 95% credible interval in brackets; an asterisk denotes an interval excluding zero. Side bias uses the full sample; betting rate, stake deviation, and welfare loss (shares of the endowment) use the confirmatory sample. The 0–6 class contains only the two pure sequences.

Table C15: Participant versus sequence heterogeneity, by outcome

	$\sigma_{\text{participant}}$	σ_{sequence}	Ratio	Participant share
Betting, logit	3.07 [2.58, 3.68]	0.80 [0.65, 1.00]	3.86 [2.97, 5.02]	0.94 [0.90, 0.96]
Welfare loss, positive-loss mean	1.33 [1.16, 1.55]	0.21 [0.16, 0.26]	6.39 [5.01, 8.31]	0.98 [0.96, 0.99]
Side choice, logit	0.51 [0.42, 0.61]	0.44 [0.35, 0.55]	1.15 [0.86, 1.53]	0.57 [0.43, 0.70]
Posterior draws	1000			

Notes: Standard deviations of the participant- and sequence-level random intercepts in the preregistered model for each outcome (RQ1 betting, RQ3 welfare loss, RQ4 side choice), on each model's linear-predictor scale. Because scales differ across models, only the within-model ratio $\sigma_{\text{participant}}/\sigma_{\text{sequence}}$ and the participant share $\sigma_{\text{participant}}^2/(\sigma_{\text{participant}}^2 + \sigma_{\text{sequence}}^2)$ are comparable across rows. Cells report posterior medians and 95% credible intervals. Side choice uses the full sample; the other outcomes the confirmatory sample.

Table C16: Participant versus sequence heterogeneity, by outcome

	$\sigma_{\text{participant}}$	σ_{sequence}	Ratio	Participant share
Betting, logit	2.53 [2.07, 3.12]	0.56 [0.41, 0.72]	4.61 [3.29, 6.70]	0.96 [0.92, 0.98]
Welfare loss, zero-loss hurdle	2.72 [2.22, 3.39]	0.59 [0.45, 0.77]	4.62 [3.28, 6.29]	0.96 [0.91, 0.98]
Welfare loss, positive-loss mean	1.83 [1.54, 2.25]	0.23 [0.18, 0.29]	7.89 [5.90, 10.72]	0.98 [0.97, 0.99]
Side choice, logit	0.34 [0.22, 0.51]	0.41 [0.28, 0.54]	0.85 [0.51, 1.39]	0.42 [0.20, 0.66]
Posterior draws	1000			

Notes: Standard deviations of the participant- and sequence-level random intercepts in the preregistered model for each outcome (RQ1 betting, RQ3 welfare loss, RQ4 side choice), on each model's linear-predictor scale. Because scales differ across models, only the within-model ratio $\sigma_{\text{participant}}/\sigma_{\text{sequence}}$ and the participant share $\sigma_{\text{participant}}^2/(\sigma_{\text{participant}}^2 + \sigma_{\text{sequence}}^2)$ are comparable across rows. Cells report posterior medians and 95% credible intervals. Side choice uses the full sample; the other outcomes the full sample.

conventional concern thresholds of 5 or 10. The side-choice specification consistently shows the highest VIF of the three, which is expected rather than a specification problem: it is the only model that uses the signed versions of imbalance and the terminal-run indicator, which are more strongly correlated by construction than their absolute-value counterparts used in the other two models (a sequence far from balanced is also more likely to end on a run in the same direction). The pattern is stable across treatments, with near-identical VIF values in m25 and m19 for each specification, indicating that the residual collinearity between the two predictors reflects a fixed feature of the sequence design rather than treatment-specific sampling variation.

Table C17: Variance Inflation Factors by Model and Treatment

Model	m25	m19
Extensive margin (bet)	1.04	1.04
Intensive margin (stake)	1.06	1.07
Side choice (heads)	1.57	1.61

C.7.2 Average Marginal Effects

Table C18: Average Marginal Effects of Imbalance and Terminal Run on Betting and Side Choice

	FN ($m = 2.5$)	FP ($m = 1.9$)
<i>Panel A: Extensive margin ($P(bet)$, $p.p.$)</i>		
Imbalance	22.6 [18.6, 26.6]***	24.7 [18.2, 31.2]***
Terminal run ≥ 2	0.1 [-1.2, 1.5]	2.4 [-0.1, 4.8] [†]
<i>Panel B: Side choice ($P(signaled\ side)$, $p.p.$)</i>		
Imbalance (signed)	11.0 [7.7, 14.2]***	8.8 [2.9, 14.6]**
Terminal run (signed)	-15.3 [-17.1, -13.4]***	-14.2 [-17.6, -10.8]***

Notes: Average marginal effects (AMEs) from logistic models described in Section ?? . Panel A uses unsigned (absolute) imbalance and terminal-run predictors and is estimated on the full sample (FN: $n = 9,536$ trial obs., 149 participants; FP: $n = 3,392$, 53). Panel B uses signed imbalance and a signed terminal-run indicator with side chosen as the outcome, estimated on the subsample of trials in which a bet was placed (FN: $n = 7,333$, 147 participants; FP: $n = 2,312$, 50). Brackets show 95% confidence intervals. [†] $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

C.8 Robustness Checks: The Betting-Rate/Side-Choice Mechanism

This section reports the full set of checks supporting the claim that the association between a sequence’s side-bias and its betting-rate shift is not an artifact of selective participation. Because side choice is only observed conditional on betting, sequences with lower betting rates have their heads-share estimated on a smaller, self-selected sample; Panel A of Table C19 asks whether this selection mechanism can account for the pattern in Table ?? and Figure ?. As expected mechanically, the width of a sequence’s heads-share credible interval is strongly related to its betting-rate shift ($\rho = -0.77$, $p < .001$): sequences with fewer bettors carry more posterior uncertainty about the side choice of those who did bet. This mechanical relationship does not, however, extend to the point estimates themselves. The *magnitude* of a sequence’s side-bias (—heads-share Δ —) is unrelated to its betting-rate shift, whether measured across all 64 sequences ($\rho = -0.00$, $p = .98$), between sequences below versus above the grand mean of betting rate ($p = .33$), within mirror pairs directly (paired Wilcoxon, $p = .64$), or restricting to only the sequences with a credible betting-rate shift ($\rho = .05$, $p = .76$). The within-pair comparison for CI width, the strictest test of the mechanical story, is itself not significant ($p = .15$), indicating that the strong aggregate relationship in column (1) is driven in part by between-pair variation rather than holding pair-by-pair.

Panel B tests whether the betting-rate shift itself is better described by two distinct sequence types (raising vs. lowering betting) than by a single continuous distribution, as suggested by the visual clustering in Figure ?. A two-component Gaussian mixture is preferred to a one-component fit ($\text{BIC}_2 - \text{BIC}_1 = 18.37$, mclust convention, higher favors two components), and the bimodality coefficient (0.568) sits just above the 0.555 rule-of-thumb threshold conventionally taken as suggestive of bimodality. Both are consistent with, though not dispositive proof of, two underlying sequence types.

C.9 Participant Engagement — Eye-tracking

Eye-tracking data were available for twelve participants from a pilot session. Five areas of interest (AOIs) were defined: the sequence of previous outcomes (*Sequences*), response

Table C19: Mechanism checks: heads-share bias, betting-rate shift, and bimodality

Check	Statistic	Value	<i>p</i> -value	<i>n</i>
<i>Panel A: Mechanical-explanation checks</i>				
CI width vs. betting-rate shift (all sequences)	Spearman ρ	−0.770	< .001	64
Bias magnitude vs. betting-rate shift (all sequences)	Spearman ρ	−0.003	.978	64
CI width vs. bias magnitude (all sequences)	Spearman ρ	−0.223	.077	64
CI width, below- vs. above-mean betting rate	Mean diff.	0.010	< .001	64
Bias magnitude, below- vs. above-mean betting rate	Mean diff.	0.016	.327	64
CI width, lower- vs. higher-betting pair member (paired)	Wilcoxon <i>V</i>	341	.154	32
Bias magnitude, lower- vs. higher-betting pair member (paired)	Wilcoxon <i>V</i>	238	.638	32
Within-pair betting-rate gap vs. CI-width gap	Spearman ρ	0.124	.499	32
Bias magnitude vs. betting-rate shift, credible shifts only	Spearman ρ	0.053	.761	35
<i>Panel B: Bimodality of the betting-rate shift</i>				
Mixture components selected (1 vs. 2)	<i>G</i>	2	–	64
BIC gap (2-component – 1-component)	Δ BIC	18.37	–	64
Bimodality coefficient	<i>BC</i>	0.568	–	64

Notes: CI width is the width of the sequence’s 95% posterior credible interval for heads-share Δ ; bias magnitude is $|\hat{\mu}_s^h - \bar{\mu}^h|$. “Betting-rate shift” refers to $\hat{\mu}_s^b - \bar{\mu}^b$. Mean-difference rows report a two-sample *t*-test comparing sequences with a below- vs. above-grand-mean betting rate. Paired rows compare the lower- and higher-betting member of each of the 32 mirror pairs directly (Wilcoxon signed-rank test). The mixture model is fit via `mclust`, which reports BIC on a higher-is-better scale, so a positive gap favors the two-component fit. The bimodality coefficient’s 0.555 threshold is the value implied by a uniform distribution and is a heuristic rather than a formal test.

options (*Choices*), the wager-size slider (*Slider*), payoff information (*Payoff*), and the task-progress indicator (*Progress*). Recording completeness varied because eye-tracking was interrupted before task completion in some sessions: eight participants had more than 80% of the task successfully recorded, while completeness for the remaining four ranged from 11% to 55%.

As an initial measure of relative visual attention, we computed total fixation duration for each AOI and expressed it as a proportion of the participant’s total fixation duration across all five AOIs. *Sequences* and *Payoff* together accounted for the majority of recorded fixation time for most participants, *Choices* generally accounted for a smaller share, and relatively little time was allocated to *Slider* and, particularly, *Progress*. There was nonetheless considerable between-participant variation: some participants attended predominantly to *Sequences*, while for others *Payoff* accounted for the largest share of fixation time.

Figure C3 provides a more detailed view, showing the distribution of individual fixation durations by AOI together with their relative fixation frequencies (violin width scaled

by the number of observations). Fixation durations were broadly similar across the five AOIs and right-skewed throughout, with most fixations concentrated at short durations and relatively few substantially longer ones. Substantially more fixations were recorded for *Sequences* and *Payoff* than for *Slider* and, especially, *Progress*. Together with the fixation-time results above, this indicates that the greater total attention devoted to *Sequences* and *Payoff* was driven primarily by these areas being fixated more frequently, rather than by individual fixations being longer.

Visual inspection of the recordings provided additional context. *Sequences* received the largest number of fixations ($N = 5,523$); this AOI occupied a relatively large area of the screen, and participants typically made multiple fixations across the individual letters of the displayed sequence, suggesting that its high total fixation time reflects repeated scanning rather than unusually long individual fixations. *Payoff* received roughly half as many fixations ($N = 2,735$), but its distribution extended further toward longer durations; inspection suggested that some participants held their gaze on the payoff display while adjusting the slider, during which the displayed payoff updated dynamically with the selected wager. *Choices* showed an intermediate pattern ($N = 1,923$ fixations): although attended to less often than *Sequences* or *Payoff*, individual fixation durations were comparable to those in the other AOIs. *Slider* received considerably fewer fixations ($N = 791$); its small spatial extent likely contributed to this, and because *Slider* and *Payoff* were spatially adjacent, minor calibration inaccuracies may have caused some boundary fixations to be assigned to *Payoff* rather than *Slider*. *Progress* received the fewest fixations ($N = 91$), which inspection suggested were typically brief and occurred near the beginning of a block.

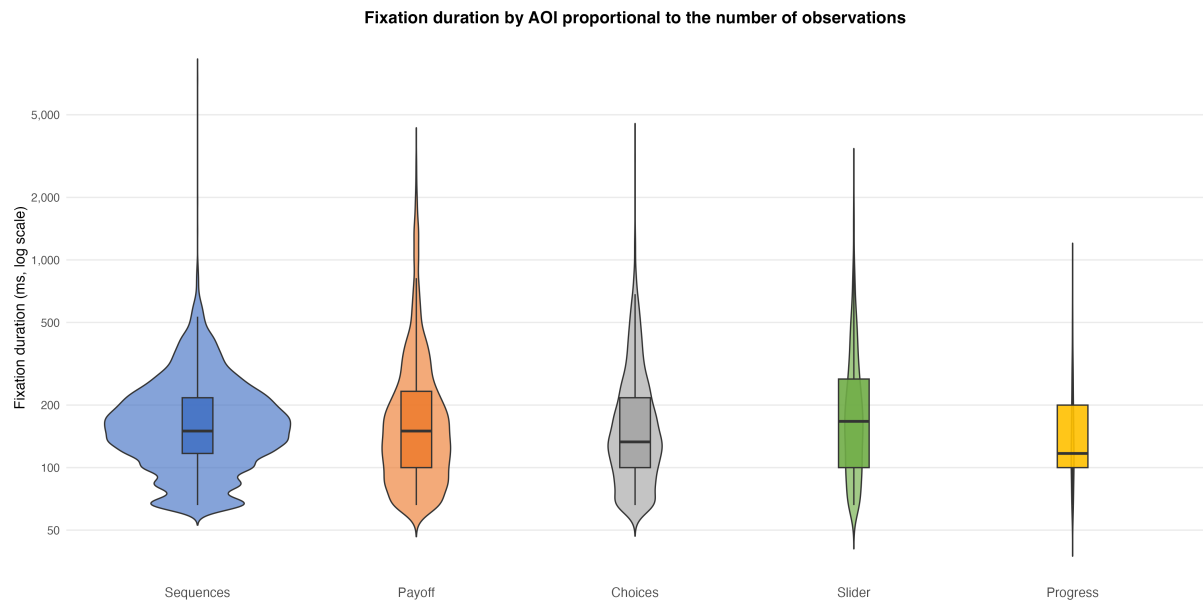


Figure C3

D APPENDIX TABLES

D.1 Stan Convergence Diagnostics

Table D1 reports the sampler settings used for each model and Table ?? reports the resulting convergence diagnostics. The preregistered convergence criteria are $\hat{R} \leq 1.01$, $\text{ESS} \geq 400$, and zero divergent transitions. All models were initially run with the default settings (2,000 iterations, 1,000 warmup, `adapt_delta` = 0.95, `max_treedepth` = 12).

Model	Iter	Warmup	Chains	<code>adapt_delta</code>	<code>max_treedepth</code>
MPL (r_i)	2000	1000	4	0.95	12
RQ1	6000	1000	4	0.97	13
RQ2	2000	1000	4	0.95	12
RQ3	8000	1000	4	0.97	13
RQ4	2000	1000	4	0.95	12
EX1	2000	1000	4	0.95	12
EX2	2000	1000	4	0.99	13

Table D1: Stan sampler settings by model. Post-warmup draws per chain = iter – warmup. RQ3 used boosted settings after the default settings failed to meet the preregistered convergence criteria for the FN treatment.

RQ1 and RQ3 were rerun with boosted settings (Table D1) to improve sampling quality. For RQ3, the boosted settings (8,000 iterations, `adapt_delta` = 0.97) brought all models within the preregistered convergence thresholds. These adjustments affect only numerical sampling performance and do not alter model likelihoods, priors, estimands, or reporting rules, consistent with the preregistered addendum. The MPL and RQ2 models met all convergence criteria under the default settings and were not rerun.

The RQ3 FN Gamma-only model meets the preregistered convergence criteria but yields moderate sequence ordering stability relative to the Gaussian alternative (Spearman $\rho \approx 0.49\text{--}0.57$). This reflects weak identification of sequence-level welfare loss effects due to the dominant role of individual heterogeneity ($\sigma_u/\sigma_s = 6.41$) rather than a computational failure. Sequence-level welfare loss ordering beyond the pure sequences should accordingly be interpreted with caution; see the main text for discussion.

D.2 Untitled

. We operationalized the preregistered boundary inflation check using a posterior predictive p-value based on the Kolmogorov-Smirnov statistic applied to the distribution of standardized stake deviations. The observed KS statistic fell well within the range of replicated values in both treatments (PPC p-value: FN = 0.71, FP = 0.24), indicating no evidence of boundary inflation. The preregistered one-inflated Beta-Binomial robustness model was therefore not fitted. As a descriptive supplement, all-in stakes constituted 17.7% of FN betting trials and 11.2% of FP betting trials.

D.3 Output Tables

All output files are available in the project OSF repository in the **XY** folder. The table below maps each data source number to its filename and description.

Slope	Covariate	Measure	FN Treatment ($n = 147$)		FP Treatment ($n = 50$)	
			ρ [95% CrI]	$P(\rho > 0)$	ρ [95% CrI]	$P(\rho > 0)$
Run	Optimism	Signed	0.05 [−0.05, 0.14]	.82	−0.03 [−0.19, 0.11]	.33
Run	Optimism	Magnitude	−0.08 [−0.19, 0.03]	.06	0.04 [−0.14, 0.23]	.66
Run	Log RT	Signed	− 0.29 [−0.38, −0.18]	.00	− 0.29 [−0.45, −0.13]	.00
Run	Log RT	Magnitude	0.25 [0.14, 0.35]	1.00	0.25 [0.05, 0.43]	.99
Run	Risk pref.	Signed	0.02 [−0.08, 0.11]	.64	0.07 [−0.08, 0.22]	.82
Run	Risk pref.	Magnitude	−0.03 [−0.13, 0.07]	.28	−0.19 [−0.37, 0.01]	.03
Imbalance	Optimism	Signed	0.00 [−0.06, 0.05]	.44	0.02 [−0.06, 0.11]	.72
Imbalance	Optimism	Magnitude	− 0.15 [−0.23, −0.07]	.00	− 0.23 [−0.36, −0.09]	.00
Imbalance	Log RT	Signed	0.04 [−0.02, 0.09]	.91	0.02 [−0.08, 0.13]	.67
Imbalance	Log RT	Magnitude	0.10 [0.02, 0.18]	.99	0.11 [−0.03, 0.25]	.94
Imbalance	Risk pref.	Signed	0.14 [0.08, 0.19]	1.00	0.05 [−0.06, 0.15]	.83
Imbalance	Risk pref.	Magnitude	− 0.11 [−0.19, −0.03]	.00	− 0.19 [−0.34, −0.04]	.01

Notes: Each row correlates a participant-level side-choice slope (run- or imbalance-slope, population fixed effect plus own random-effect draw) against a fixed individual-difference covariate (dispositional optimism, mean log response time on betting trials, or CRRA risk preference r), computed within each posterior draw and summarized by posterior median, 95% credible interval, and $P(\rho > 0)$. *Signed* rows use the slope as estimated (sign indicates direction: negative run-slope = against the terminal run, positive imbalance-slope = with the majority). *Magnitude* rows use the absolute value of the slope, a direction-agnostic measure of how strongly a participant relies on that cue, regardless of which way they lean. Bold entries denote a 95% credible interval excluding zero. n is the number of participants with non-missing values on all three covariates in that treatment. The signed imbalance-slope is credibly, positively associated with risk preference in FN ($\rho = 0.14$ [0.08, 0.19]) but not FP, suggesting risk-averse participants who do respond to the imbalance cue lean toward the majority side—this direction-specific result does not replicate across treatments and should be read as suggestive rather than confirmed. The run and imbalance cues are themselves moderately correlated across the 64 distinct sequences ($r = 0.58$ between the terminal-run indicator and signed imbalance), so the magnitude comparisons above are not fully independent tests.

Table D5: Correlations between participants’ personal side-choice slopes and individual-difference covariates.

Table D6: Welfare loss, betting, and staking by slope class

	n_A	n_B	Group A	Group B	Difference (A – B)
<i>Panel A: Welfare loss (% of endowment)</i>					
Against imbalance vs. uncertain on imbalance	21	46	7.3	17.6	–10.0* [–19.8, –0.4]
Against imbalance vs. with imbalance	21	28	7.3	8.0	–0.7 [–6.3, 7.0]
Against imbalance vs. all others	21	74	7.3	13.9	–6.3 [–13.0, 1.8]
Against run vs. uncertain on run	36	58	11.8	12.9	–0.8 [–9.6, 7.9]
Uncertain on both vs. all others	31	64	18.9	9.3	9.4* [1.2, 23.2]
<i>Panel B: Betting rate (%)</i>					
Against imbalance vs. uncertain on imbalance	21	46	81.5	80.7	0.8 [–10.4, 12.4]
Against imbalance vs. with imbalance	21	28	81.5	67.2	14.4* [2.6, 25.8]
Against imbalance vs. all others	21	74	81.5	75.2	6.2 [–4.4, 16.4]
Against run vs. uncertain on run	36	58	78.0	75.6	2.4 [–8.7, 12.2]
Uncertain on both vs. all others	31	64	78.1	76.3	1.8 [–11.2, 12.2]
<i>Panel C: Absolute stake deviation (pp)</i>					
Against imbalance vs. uncertain on imbalance	21	46	18.7	31.3	–12.6* [–22.1, –2.6]
Against imbalance vs. with imbalance	21	28	18.7	24.5	–5.7 [–14.8, 3.7]
Against imbalance vs. all others	21	74	18.7	28.8	–10.0* [–18.0, –1.3]
Against run vs. uncertain on run	36	58	23.7	28.1	–4.3 [–13.3, 5.2]
Uncertain on both vs. all others	31	64	33.5	23.1	10.5* [2.0, 21.2]

Note: Participants are classified as *against*, *with*, or *uncertain* by whether the 95% credible interval of their personal side-choice slope on imbalance (or on the terminal run) lies below zero, above zero, or includes zero. Groups A and B are the two sides of each comparison; cells report group means, and the difference is A – B with 95% credible intervals in brackets. Welfare loss, betting rate, and stake deviation are participant-level posterior means from the RQ3, RQ1, and RQ2 models. Each statistic is computed within 1,000 posterior draws and combined with Bayesian-bootstrap weights over participants, so intervals reflect estimation and sampling uncertainty. An asterisk denotes an interval excluding zero. Confirmatory sample ($n = 95$).

Model	Treatment	Subset (N)	Likelihood	$\hat{R}_{\max}$	ESS _{min}	Div.	PPC p	Selected
<i>Preregistered convergence criteria</i>				≤ 1.01	≥ 400	0		
MPL (r_i)	Pooled	Full	logistic CRRA	1.003	1439	0	–	
	FN	Full	logistic CRRA	1.002	1543	0	–	
	FP	Full	logistic CRRA	1.004	1511	0	–	
	Pooled	Consistent	logistic CRRA	1.005	1517	0	–	
	FN	Consistent	logistic CRRA	1.003	1526	0	–	
	FP	Consistent	logistic CRRA	1.005	1645	0	–	
RQ1	FN	Confirmatory	Bernoulli ^P	1.002	1164	0	.45	✓
			Beta-Binomial ^a	1.002	1190	0	–	
	FP	Full	Bernoulli ^P	1.006	1035	0	.47	✓
			Beta-Binomial ^a	1.003	1269	0	–	
			Bernoulli ^P	1.001	906	0	.51	✓
			Beta-Binomial ^a	1.002	943	0	–	
RQ2	FN	Confirmatory	Gaussian ^P	1.004	1223	0	.79	✓
			Beta-Binomial ^a	1.011	185	0	–	
	FP	Full	Gaussian ^P	1.002	1105	0	.52	✓
			Beta-Binomial ^a	1.029	115	0	–	
			Gaussian ^P	1.003	1097	0	.18	✓
			Beta-Binomial ^a	1.013	232	0	–	
RQ3	FN	Confirmatory	hurdle-Gamma ^P	1.002	599	0	–	
			Gamma-only ^s	1.008	741	0	.49 [†]	✓
			Gaussian ^a	1.004	394	0	–	
	FP	Full	hurdle-Gamma ^P	1.011	495	0	–	
			Gamma-only ^s	1.006	590	0	.57 [†]	✓
			Gaussian ^a	1.003	516	0	–	
	FN	Full	hurdle-Gamma ^P	1.009	1004	0	.91 [†]	✓
			Gamma-only ^s	1.002	1355	0	–	
RQ4	FN	Full ($N = 147$)	Bernoulli ^P	1.003	726	0	.48 [‡]	✓
	FP	Full ($N = 50$)	Bernoulli ^P	1.009	941	0	.46 [‡]	✓
EX1	FN	Full	Gaussian (MI)	1.00X	XXXX	0	–	✓
	FP	Full	Gaussian (MI)	1.00X	XXXX	0	–	✓
EX2	FN	Full	Gaussian (MI)	1.006	973	0	–	✓
	FP	Full	Gaussian (MI)	1.003	1283	0	–	✓

Table D2: Stan convergence diagnostics for all fitted models. $\hat{R}_{\max}$ and ESS_{min} are the worst-case values across all parameters. Div. denotes the number of divergent transitions. The PPC column reports the posterior predictive p -value for RQ1 and RQ2 (values near 0.5 indicate good fit); for RQ3 it reports the Spearman ρ between the selected model and the Gaussian alternative as a sequence ordering stability check ([†]), since the PPC p -value is not computed for that model; for RQ4 it reports the overdispersion p -value $\Pr(\text{Var}_{\text{rep}} \geq \text{Var}_{\text{obs}})$, where Var is the variance of sequence-wise Heads rates ([‡]); values near 0.5 indicate the Bernoulli model adequately captures between-sequence dispersion, and values below 0.05 would trigger the Beta-Binomial robustness fit. Likelihood superscripts: ^P preregistered primary; ^s preregistered secondary; ^a diagnostic alternative. The preregistered convergence criteria are $\hat{R} \leq 1.01$ and $\text{ESS} \geq 400$; values violating these thresholds are shown in bold. All selected models meet the convergence criteria. The FN full-sample RQ1 model includes a linear drift term; all other RQ1 models use no drift adjustment. The diagnostic alternative models for RQ2 fail the ESS criterion and are not selected; they are reported only for completeness. For RQ3 FN, the selected Gamma-only model meets the convergence criteria but yields moderate sequence ordering stability ($\rho \approx 0.49$ – 0.57), reflecting weak identification of sequence-level effects relative to participant heterogeneity rather than a computational failure; see main text for discussion. The MPL model estimates individual CRRA risk parameters r_i from the Holt–Laury task and meets all convergence criteria across all subsets. Sample sizes for RQ4: $N = 147$ (FN) and $N = 50$ (FP), reflecting all participants with at least one betting trial; no normative-better or HL-consistency restriction is applied (see Section ??).

	Median Δ	95% CI	$P(\Delta > 0)$	$P(\Delta > \tau)$
<i>RQ1: Extensive margin (betting probability)</i>				
Participant level	+0.09	[0.07, 0.10]	1.00	0.04
Spearman ρ	+0.54	[0.39, 0.67]	1.00	—
Sequence range	[-0.00, +0.19]	34/64 sequences with $P(\Delta > 0) \geq 0.95$		
<i>RQ2: Intensive margin (stake deviation, conditional on betting)</i>				
Participant level	-0.26	[-0.27, -0.26]	0.00	1.00
Spearman ρ	+0.44	[0.26, 0.59]	1.00	—
Sequence range	[-0.31, -0.22]	64/64 sequences with $P(\Delta > \tau) = 1.00$		
<i>RQ3: Welfare loss (certainty-equivalent, share of endowment)</i>				
Participant level	-0.02	[-0.03, -0.01]	0.00	0.01
Spearman ρ	+0.34	[0.17, 0.51]	1.00	—
Sequence range	[-0.18, +0.03]	3/64 sequences with $P(\Delta > \tau) \geq 0.50$		

Table D3: EX3: FN–FP posterior contrasts by research question. Δ = posterior median difference (FN minus FP); $\tau = 0.10$ for RQ1, $\tau = 0.05$ for RQ2–3. $n_{\text{FN}} = 149$ (146 for RQ2), $n_{\text{FP}} = 53$ (50 for RQ2). RQ1: FN participants bet more overall, but the absolute gap rarely exceeds τ at the participant level; substantial heterogeneity exists across sequences. RQ2: the FN–FP gap is uniform and negative across all sequences, reflecting the mechanical consequence of the EU-optimal stake being virtually zero in FP. RQ3: aggregate welfare losses are negligible and comparable across treatments.

Table D4: **NOT REFERENCED** EX2: Participant-level associations of optimism and response time with behavioral outcomes. Posterior medians with 95% credible intervals in brackets. $P(\cdot > 0)$ is the posterior probability that the coefficient is positive. Outcomes: $\hat{\mu}_Z^b$ = betting probability (RQ1); $\hat{\mu}_Z^a$ = stake deviation (RQ2); $\hat{\mu}_Z^c$ = welfare loss (RQ3); $\hat{\mu}_Z^h$ = heads bias (RQ4). All outcomes are standardised within each posterior draw before entering the regression. Predictors: Z^{opt} = standardised LOT-R score; Z^{rt} = standardised log mean response time. Partial pooling across outcomes via non-centred hierarchical prior on slopes; τ parameters capture between-outcome heterogeneity. MI likelihood marginalises over 200 upstream posterior draws per participant.

Outcome	Parameter	FN ($m = 2.5$)		FP ($m = 1.9$)	
		Median [95% CI]	$P(\cdot > 0)$	Median [95% CI]	$P(\cdot > 0)$
<i>Pooled slopes</i>					
	$\bar{\beta}^{\text{opt}}$	0.05 [−0.10, 0.18]	0.82	−0.06 [−0.29, 0.20]	0.29
	$\bar{\beta}^{\text{rt}}$	−0.15 [−0.34, 0.06]	0.05	−0.30 [−0.50, −0.09]	0.01
<i>Between-outcome heterogeneity</i>					
	τ^{opt}	0.05 [0.00, 0.32]	—	0.12 [0.01, 0.52]	—
	τ^{rt}	0.12 [0.01, 0.46]	—	0.08 [0.00, 0.42]	—
<i>Outcome-specific slopes: optimism (Z^{opt})</i>					
$\hat{\mu}_Z^b$	β^{opt}	0.05 [−0.07, 0.16]	0.81	0.01 [−0.19, 0.25]	0.52
$\hat{\mu}_Z^a$	β^{opt}	0.04 [−0.09, 0.15]	0.77	−0.10 [−0.34, 0.10]	0.16
$\hat{\mu}_Z^c$	β^{opt}	0.07 [−0.03, 0.19]	0.92	−0.03 [−0.22, 0.19]	0.39
$\hat{\mu}_Z^h$	β^{opt}	0.04 [−0.09, 0.15]	0.75	−0.11 [−0.34, 0.07]	0.13
<i>Outcome-specific slopes: response time (Z^{rt})</i>					
$\hat{\mu}_Z^b$	β^{rt}	−0.21 [−0.36, −0.09]	0.00	−0.34 [−0.56, −0.16]	0.00
$\hat{\mu}_Z^a$	β^{rt}	−0.15 [−0.29, −0.02]	0.01	−0.30 [−0.49, −0.10]	0.00
$\hat{\mu}_Z^c$	β^{rt}	−0.18 [−0.31, −0.06]	0.00	−0.29 [−0.48, −0.09]	0.00
$\hat{\mu}_Z^h$	β^{rt}	−0.05 [−0.19, 0.12]	0.27	−0.29 [−0.46, −0.10]	0.00
<i>Residual SD</i>					
$\hat{\mu}_Z^b$	σ	0.94 [0.84, 1.06]	—	0.91 [0.74, 1.13]	—
$\hat{\mu}_Z^a$	σ	0.99 [0.89, 1.11]	—	0.96 [0.80, 1.20]	—
$\hat{\mu}_Z^c$	σ	0.83 [0.74, 0.95]	—	0.89 [0.71, 1.12]	—
$\hat{\mu}_Z^h$	σ	0.78 [0.67, 0.91]	—	0.58 [0.42, 0.81]	—

D.4 Sequences Overview

Table with sequences and their effects accross research questions.

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=== TABLE ===

=====

E APPENDIX FIGURES

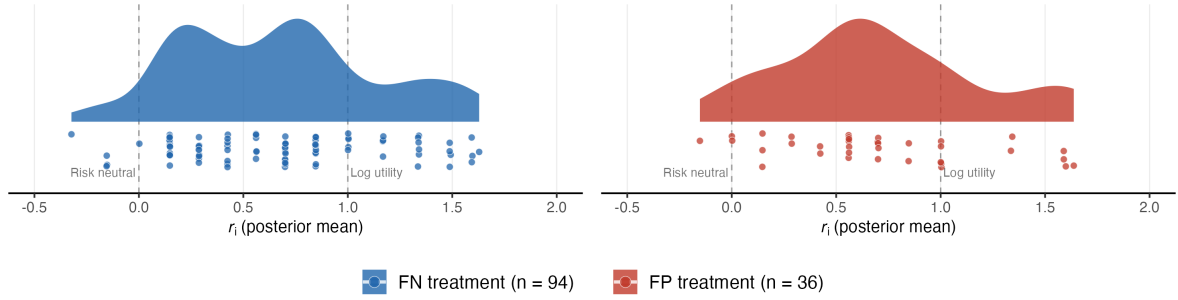


Figure E1: Pooled posterior distribution of individual risk-aversion parameters (r_i , posterior mean) by treatment, restricted to participants with consistent Holt–Laury responses. *Left panel:* FN treatment ($m = 2.5$, $n = 94$). *Right panel:* FP treatment ($m = 1.9$, $n = 36$). Each point is one participant’s posterior mean r_i ; the density above summarizes its distribution within the treatment. Dashed vertical lines mark $r_i = 0$ (risk neutrality) and $r_i = 1$ (log utility). The distribution spans a wide range in both treatments, from near risk-neutral to strongly risk-averse preferences — a spread that later proves consequential for individual-level stake calibration (Section ??).

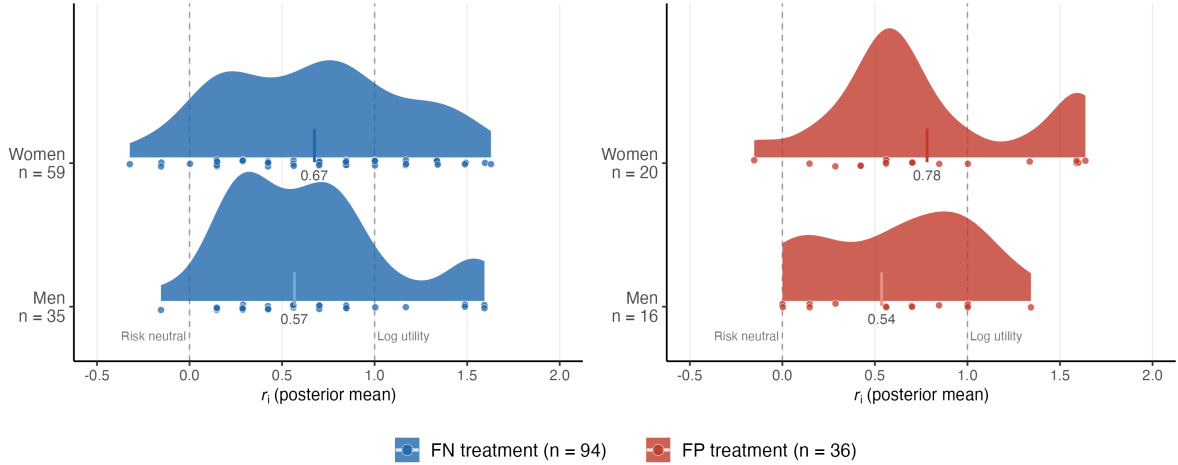


Figure E2: Distribution of individual CRRA risk-aversion parameters (r_i) by treatment and sex, restricted to participants with consistent Holt–Laury switching patterns. Each point is the posterior mean of r_i ; shaded areas show kernel density estimates. Dashed vertical lines mark risk neutrality ($r = 0$) and log utility ($r = 1$).

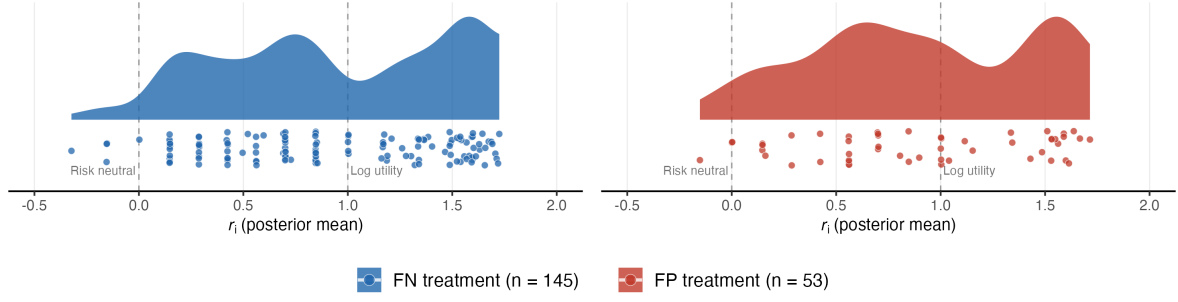


Figure E3: **not cited** Distribution of individual CRRA risk-aversion parameters (r_i) by treatment, full sample. Each point is the posterior mean of r_i ; shaded areas show kernel density estimates. Dashed vertical lines mark risk neutrality ($r = 0$) and log utility ($r = 1$).

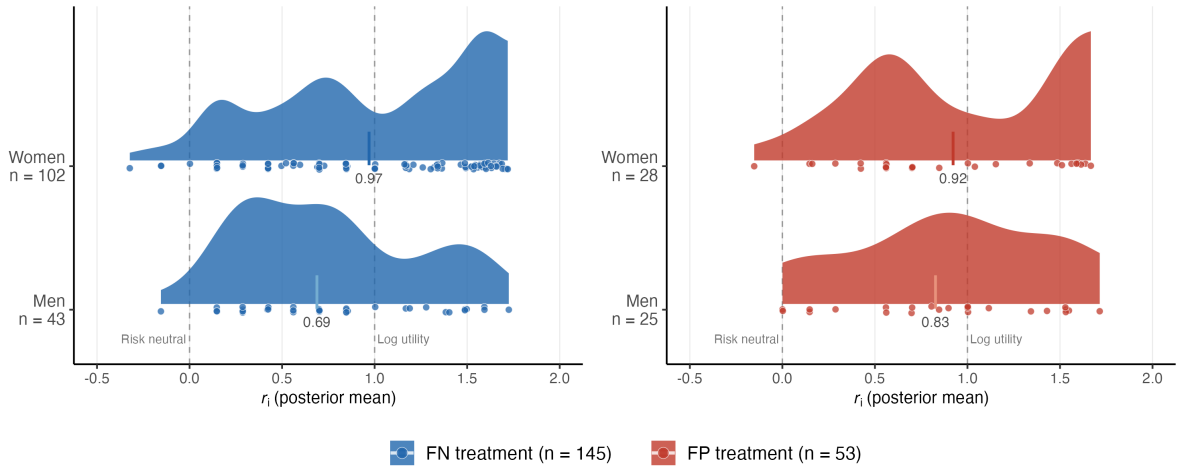


Figure E4: **not cited** Distribution of individual CRRA risk-aversion parameters (r_i) by treatment and sex, full sample. Each point is the posterior mean of r_i ; shaded areas show kernel density estimates. Dashed vertical lines mark risk neutrality ($r = 0$) and log utility ($r = 1$).

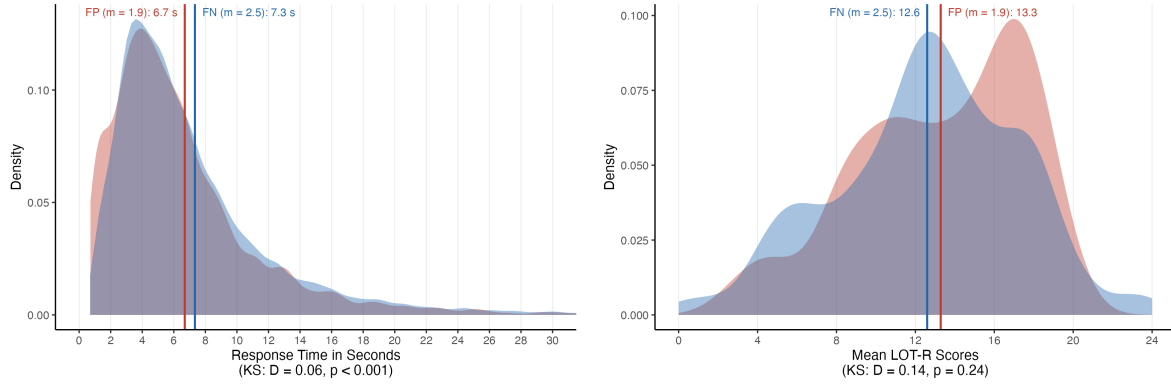


Figure E5: Distribution of response times and dispositional optimism by treatment. *Left panel*: sequence-level response times across all decisions; the x-axis is truncated at 30 seconds for readability (153 observations with longer response times are excluded from display but included in the density estimation and Kolmogorov–Smirnov test). *Right panel*: participant-level LOT-R scores (0–24). Vertical lines mark treatment means. Response times include all participants and sequences (FN: $n = 9,536$; FP: $n = 3,392$); LOT-R includes all participants (FN: $n = 149$; FP: $n = 53$). Kolmogorov–Smirnov test statistics are shown in axis labels. For response times, $D = 0.06$ indicates a negligible practical difference despite statistical significance ($p < 0.001$), driven by the large number of sequence-level observations; p-values are approximate due to ties in the timestamp data. For LOT-R, no significant distributional difference is detected ($D = 0.14$, $p = 0.24$).

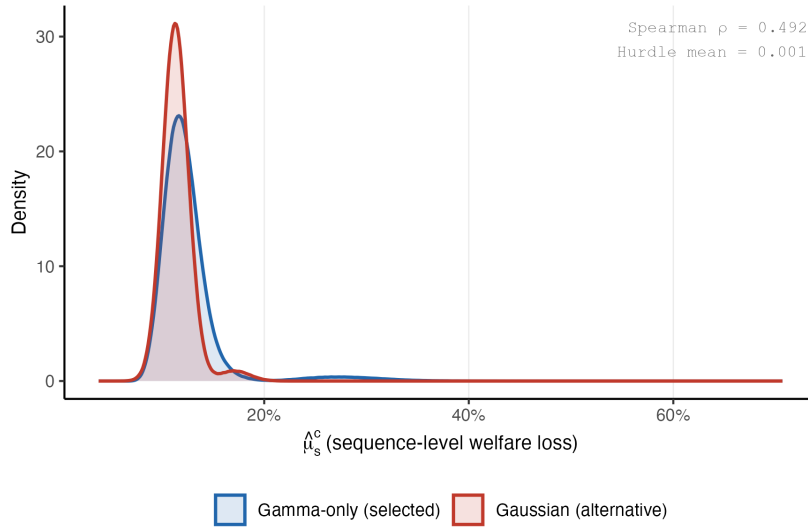


Figure E6: Posterior predictive comparison for RQ3 (welfare loss, FN treatment, confirmatory subset). Each curve shows the marginal posterior distribution of sequence-level mean welfare loss $\hat{\mu}_s^c$ across all 64 sequences under the Gamma-only model (black, selected) and the Gaussian alternative (red). The two specifications agree closely in the right tail—the high-welfare-loss sequences that drive the main findings—but diverge near the peak, where the Gaussian places a sharper and taller mode than the Gamma-only. This pattern explains the moderate Spearman rank correlation ($\rho = 0.49$): the disagreement is concentrated in the dense cluster of near-average sequences, where ordering uncertainty is least consequential, rather than among the sequences with the largest welfare losses. Hurdle probability $\bar{\omega} = 0.001$, confirming that the hurdle component is negligible and the Gamma-only specification is appropriate.

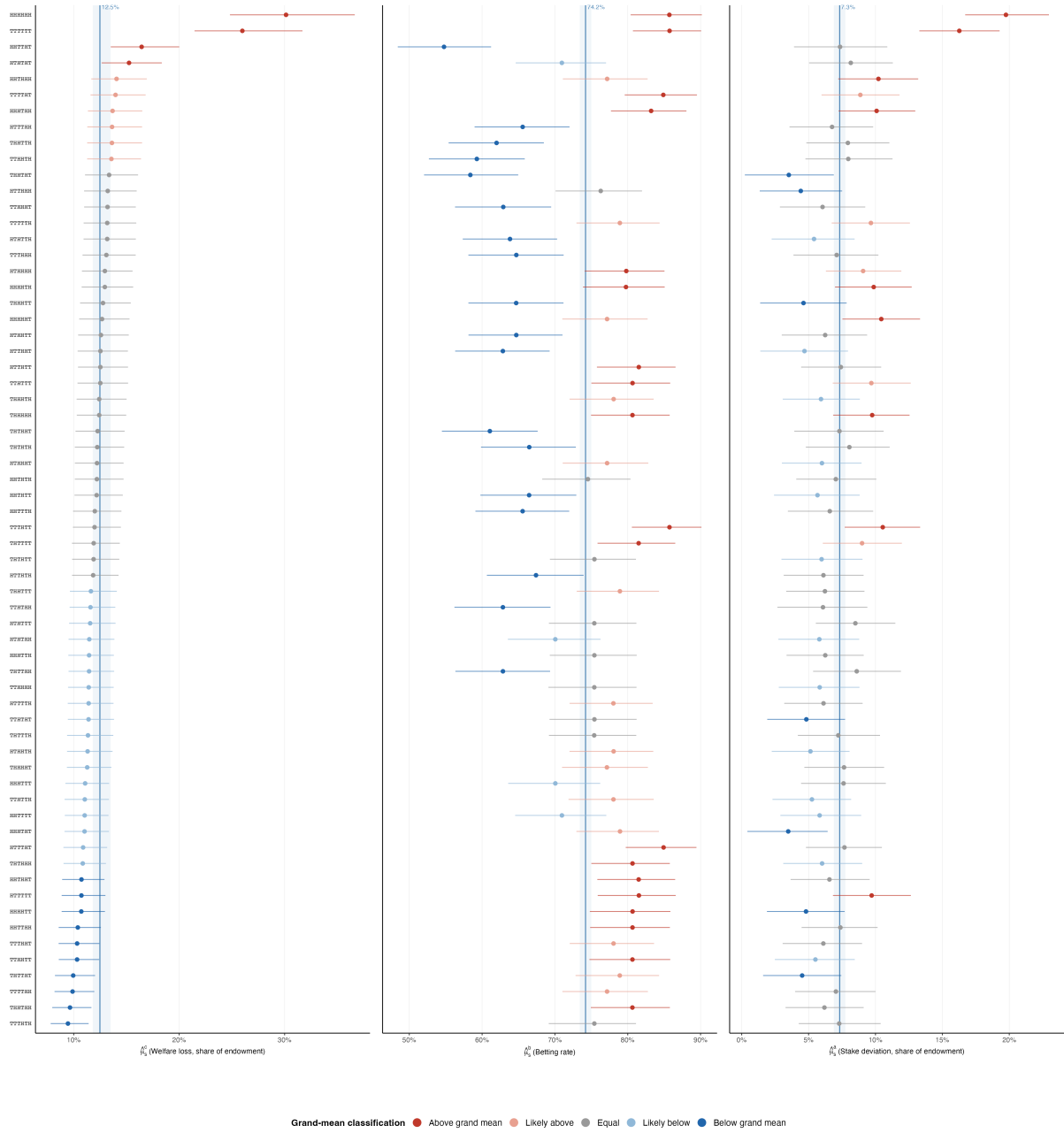


Figure E7: Sequence-level posterior estimates for RQ1–3, ordered by posterior mean welfare loss (RQ3, top = highest loss). Each row corresponds to one of the 64 six-toss sequences. Points show posterior means; horizontal lines show 95% credible intervals. The vertical line and shaded band mark the grand mean and its 95% credible interval. Sequence labels are colored by the number of research questions in which the sequence is classified as credibly above or below the grand mean: red = two, navy = one, gray = none.

	Direction	N	%	$\bar{\mu}$	Med.	Min.	Max.
<i>Head-biased</i>	Strong	4	6.2	0.68	0.67	0.64	0.73
	Moderate	11	17.2	0.61	0.61	0.60	0.63
	Weak	5	7.8	0.58	0.58	0.56	0.59
<i>Neutral</i>		24	37.5	0.51	0.51	0.46	0.55
<i>Tail-biased</i>	Strong	6	9.4	0.36	0.37	0.31	0.39
	Moderate	7	10.9	0.41	0.42	0.39	0.42
	Weak	7	10.9	0.44	0.44	0.43	0.46
<i>Total</i>		64	100				

Table E1: Classification of the 64 sequences by directional bias in Heads choice probability (FN treatment, $N = 147$). Direction is based on $\Pr(\mu_h^s \geq \bar{h} \pm \delta)$ with $\delta = 0.05$: *strong* (> 0.95), *moderate* (> 0.80), *weak* (> 0.50), *neutral* (neither direction reaches 0.50). $\bar{\mu}$, Med., Min., Max. are summary statistics of the posterior mean μ_h^s within each category.

E.1 Decision Screens

Vaše rozhodnutí pro hod mincí následující po zobrazené sekvenci:

Orel

Hlava

Nesázet

Pokračovat

Figure E8

Vaše rozhodnutí pro hod mincí následující po zobrazené sekvenci:

Orel

Hlava

Nesázet

Kliknutím na posuvník zvolte výši sázky (1—100 ECU).

Sázka: – / Orel

Pokračovat

Figure E9

Vaše rozhodnutí pro hod mincí následující po zobrazené sekvenci:

Orel

Hlava

Nesázet

Kliknutím na posuvník zvolte výši sázky (1—100 ECU).

Sázka: 50 ECU / Orel

Bude-li toto kolo vybráno k vyplacení a padne zvolená strana, obdržíte 175 ECU.
Pokud padne druhá strana, obdržíte 50 ECU.

Pokračovat

Figure E10

Vaše rozhodnutí pro hod mincí následující po zobrazené sekvenci:

Orel	Hlava	Nesázet
------	-------	---------

Bude-li toto kolo vybráno k vyplacení, získáte odměnu 100 ECU.

Pokračovat

Figure E11

F EXPERIMENTAL INSTRUCTIONS

F.1 English Translation

F.1.1 Screen 1 — General Instructions

General information

This study consists of two incentivised tasks and one questionnaire. Detailed instructions will be displayed at the beginning of each part. The expected duration is approximately 30 minutes. The study is administered by Vojtěch Zíka, Ph.D. (vojtech.zika@ujep.cz).

Compensation

For participating in this study you will receive a financial reward consisting of a fixed and a variable component.

Fixed component. As a participant in this study you are entitled to a payment of 150 CZK. This amount does not depend on your decisions during the study.

Variable component. The amount and calculation of the variable earnings will be described in detail at the beginning of each part. Variable earnings are denominated in experimental currency units (ECU). The conversion rate between ECU and Czech crowns is 1 ECU = 1 CZK. Variable earnings depend on your decisions during the study. Please read the instructions carefully and follow them; inattentive or random responding may reduce your variable earnings. Payment will be made after the session according to the administrator's instructions.

Informed consent

To proceed with the study, we need to obtain your informed consent. Please read the following information carefully.

Purpose of the study and data anonymity. This study is conducted to collect data for academic research on human behaviour and decision-making. The collected data are anonymous and cannot be linked back to any individual. Data may be used in a

research article that could be published in an academic journal. In accordance with open-science principles, anonymised raw data may be made publicly available on designated platforms (e.g., osf.io). Published data will not contain any information that could identify participants.

Content and demands of the study. The study consists of economic games and questionnaires. The demands of any part of this study do not exceed the physical or mental demands of ordinary office work.

Conduct during the study. Do not communicate with other participants and follow the instructions of the administrators. Unless necessary, please refrain from eating and drinking during the study. Please set all electronic devices to a mode in which they will not disturb you or other participants. If you have any questions or encounter a technical problem, raise your hand and wait quietly for an administrator.

Voluntariness and early withdrawal. Participation is voluntary and you may withdraw at any time without giving a reason. If you wish to withdraw early, raise your hand and wait for an administrator. In the event of early withdrawal you are entitled to the fixed participation fee but not to any earnings accrued during the study itself.

Consent to participate. Do you agree to the conditions of participation? By continuing you confirm that you have read and understood the instructions.

☐ Yes ☐ No

F.1.2 Screen 2 — Demographic Questions

Demographic questions

The following questions are used exclusively for analytical purposes. In the *Age* field, enter your age in completed years. In the *Sex* field, indicate the sex assigned at birth; if you do not wish to provide this information, select “Prefer not to say”.

Age: _____

Sex: Female / Male / Prefer not to say

F.1.3 Screen 3 — Part I (Instructions)

This part of the study consists of 64 rounds in which you may bet on the outcome of a toss of a fair coin, where Heads and Tails are equally likely.

In each round you will see a sequence of six previous tosses of such a coin. These sequences were generated in advance and serve as model scenarios.

The rounds are divided equally into 4 blocks separated by short breaks. The symbols H (Heads) and T (Tails) within each sequence are displayed from the oldest toss on the left to the most recent toss on the right.

Decision. The decision you make in each round concerns a bet on the toss that follows the displayed sequence.

Betting is conducted through a simple interface displaying three buttons: *Heads*, *Tails*, and *No bet*:

- If you wish to bet, select Heads / Tails and use the slider to set the amount you wish to stake on the chosen side.
- If you do not wish to bet, select *No bet* and proceed to the next round without placing a stake.
- The positions of the *Heads* and *Tails* buttons may change between rounds.

The decisions you make in each round affect your earnings from this part of the study.

Earnings. At the beginning of each round you are endowed with 100 ECU. In each round you may stake any amount on one side of the coin (Heads / Tails), or choose No bet. Earnings are calculated according to the same rule in every round:

- The unstaked amount is retained regardless of which side of the coin lands in that round.
- A winning stake is multiplied by 2.5. A losing stake is forfeited.

Your unstaked amount is therefore always retained, while your staked amount is either multiplied or lost.

At the end of the study, one of the 64 completed rounds will be drawn at random. The decision you made in that round determines your earnings from this part of the study. If you placed a bet in that round, the coin will be tossed — if the side you chose lands, your bet is successful.

The amount earned in the drawn round will be paid out at the rate of 1 ECU = 1 CZK.

Your earnings from this part of the study are therefore determined by decisions from one round only. Because you do not know in advance which round that will be, it is optimal to decide in each round as if that round were the one to be paid.

Practice round and comprehension check. After clicking the button, a practice round will appear showing the layout of the decision screen as it will look in the actual task. Individual elements of the screen will be briefly described in a red information box at the top of the page.

The practice round is for familiarisation only and does not affect your earnings. After completing it, a short comprehension quiz will appear to verify your understanding of the key aspects of this part of the study.

F.1.4 Screen 4 — Part I (Practice Round)

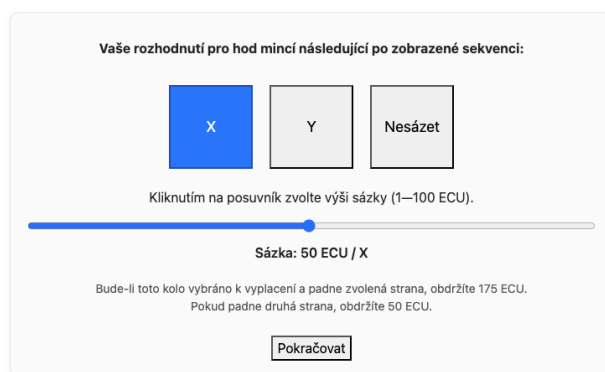
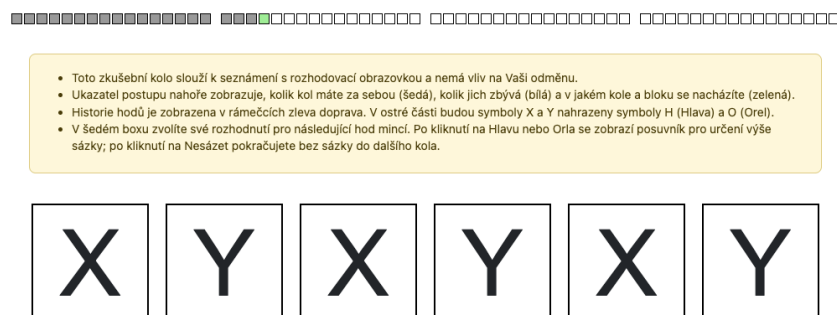


Figure F1

F.1.5 Screen 5 — Part I (Comprehension Check)

Comprehension check. The following questions let you verify that you understand the key instructions for this part of the study. If you are unsure, you may display the instructions again at any time.

[Display instructions]

1. If in the round drawn for payment you choose *No bet*, your earnings will be:

- ☐ 0 ECU
- ☐ 100 ECU
- ☐ 250 ECU

2. If in the round drawn for payment you place a bet and the bet is successful, your earnings will be:
- ☐ 100 ECU
 - ☐ The unstaked amount
 - ☐ 2.5 times the staked amount
 - ☐ The unstaked amount plus 2.5 times the staked amount
3. A bet is successful if the side you chose matches the side that:
- ☐ is most frequent in the displayed sequence
 - ☐ is the last outcome in the displayed sequence
 - ☐ lands on the toss of a fair coin following the displayed sequence
4. Earnings from this part of the study will be determined by decisions:
- ☐ in one randomly selected round
 - ☐ in all rounds
 - ☐ in the last round
5. Which symbol (H = Heads, T = Tails) is the most recent (last) in the displayed sequence?
- ☐ The rightmost
 - ☐ The leftmost

F.1.6 Screens 6–72 — Part I (Sequence Betting Task)

Screens 22, 39, and 56 are between-block rest screens:

You have completed Block 1 of 4.

You will continue in 15 seconds.

F.1.7 Screen 73 — Part II (Holt–Laury Task)

Your task in this part of the study is to make 19 decisions between Option A and Option B in the table below.

Earnings from this part will be determined in two steps. First, one of your 19 decisions will be drawn at random. Then, for the option you selected in that decision, it will be determined whether you receive the higher or lower payoff, as follows:

- You receive the *higher payoff* (20 ECU for Option A and 38.5 ECU for Option B) with the stated probability, which varies across decisions (e.g., in the first decision the probability is 5% for both options, in the second 10%, etc.).
- If you do not receive the higher payoff, you receive the *lower payoff* (16 ECU for Option A and 1 ECU for Option B).

The payoff determined in this way will be paid out at the rate of 1 ECU = 1 CZK.

Summary: Make 19 decisions between Option A and Option B. In each decision, choose the option you prefer given the combination of payoff and probability.

Option A	Option B
20 ECU with probability 5%, otherwise 16 ECU	38.5 ECU with probability 5%, otherwise 1 ECU
20 ECU with probability 10%, otherwise 16 ECU	38.5 ECU with probability 10%, otherwise 1 ECU
20 ECU with probability 15%, otherwise 16 ECU	38.5 ECU with probability 15%, otherwise 1 ECU
20 ECU with probability 20%, otherwise 16 ECU	38.5 ECU with probability 20%, otherwise 1 ECU
20 ECU with probability 25%, otherwise 16 ECU	38.5 ECU with probability 25%, otherwise 1 ECU
20 ECU with probability 30%, otherwise 16 ECU	38.5 ECU with probability 30%, otherwise 1 ECU
20 ECU with probability 35%, otherwise 16 ECU	38.5 ECU with probability 35%, otherwise 1 ECU
20 ECU with probability 40%, otherwise 16 ECU	38.5 ECU with probability 40%, otherwise 1 ECU
20 ECU with probability 45%, otherwise 16 ECU	38.5 ECU with probability 45%, otherwise 1 ECU
20 ECU with probability 50%, otherwise 16 ECU	38.5 ECU with probability 50%, otherwise 1 ECU
20 ECU with probability 55%, otherwise 16 ECU	38.5 ECU with probability 55%, otherwise 1 ECU
20 ECU with probability 60%, otherwise 16 ECU	38.5 ECU with probability 60%, otherwise 1 ECU
20 ECU with probability 65%, otherwise 16 ECU	38.5 ECU with probability 65%, otherwise 1 ECU
20 ECU with probability 70%, otherwise 16 ECU	38.5 ECU with probability 70%, otherwise 1 ECU
20 ECU with probability 75%, otherwise 16 ECU	38.5 ECU with probability 75%, otherwise 1 ECU
20 ECU with probability 80%, otherwise 16 ECU	38.5 ECU with probability 80%, otherwise 1 ECU
20 ECU with probability 85%, otherwise 16 ECU	38.5 ECU with probability 85%, otherwise 1 ECU
20 ECU with probability 90%, otherwise 16 ECU	38.5 ECU with probability 90%, otherwise 1 ECU
20 ECU with probability 95%, otherwise 16 ECU	38.5 ECU with probability 95%, otherwise 1 ECU

F.1.8 Screen 74 — Part III (Life Orientation Test-Revised)

Please answer the following questions about yourself. Indicate your level of agreement using the scale below (1 = strongly disagree; 5 = strongly agree).

1. In uncertain times, I usually expect the best.
2. It is easy for me to relax.
3. If something can go wrong for me, it will.
4. I always look on the bright side of the future.
5. I enjoy my friends a lot.
6. It is important for me to keep busy.
7. I hardly ever expect things to go my way.
8. I don't get upset too easily.
9. I rarely count on good things happening to me.
10. Overall, I expect more good things to happen to me than bad.

(The following scale was displayed below each item:)

- ☐ strongly disagree
- ☐ disagree
- ☐ neither agree nor disagree
- ☐ agree
- ☐ strongly agree

F.1.9 Screen 75 — Earnings

Your earnings

Fixed participation fee:	150.00 CZK
Variable earnings:	219.00 CZK
Sequence task (drawn round: 36), earnings	180.00 CZK
Lottery choice (drawn row: 17), earnings	39.00 CZK
TOTAL EARNINGS:	369.00 CZK

F.1.10 Screen 76 — Post-Task Questionnaire

The following questions will help us better understand your decision-making during the sequence-betting task. Your answers have no effect on your earnings.

1. If you were to play 100 rounds of the sequence task, in how many rounds do you think you would win? (0 = none; 50 = half of the rounds; 100 = every round)
2. How risky did you find the betting in the sequence task? (0 = not at all; 10 = very)
3. To what extent did you believe that each toss had a 50% probability of Heads and a 50% probability of Tails regardless of the six-toss history shown? (0 = not at all; 10 = completely)
4. To what extent did you take into account the six-toss history when making your decision, even though you knew the coin was fair? (0 = not at all; 10 = very strongly)
5. How often did you bet mainly because doing “something” felt better than not betting? (0 = never; 10 = very often)
6. How bored or tired did you feel during the task? (0 = not at all; 10 = very)
7. How would you describe your overall willingness to take financial risks? (0 = very unwilling to take risks; 10 = very willing to take risks)

8. Did you use any rule or strategy when making your decisions?

☐ Yes

☐ No

☐ Not sure

9. If yes, please briefly describe it: *(optional)*

10. Do you have any further comments about this part of the study? *(optional)*

F.1.11 Screen 77 — End of Study

Thank you for your participation!

Your earnings from this study will be paid by bank transfer. Please make sure that you now have:

- A QR payment code ready. The administrator will fill in the amount. If you do not know how to do this, please have your bank account number ready instead.
- The numbered ball you drew upon entering the laboratory, which identifies your seat number.

Once you have your QR code (or account number) ready and are holding your ball, please raise your hand and wait for the administrator's instructions.

F.2 Original Czech

F.2.1 Obrazovka 1 — Obecné Instrukce

Obecné informace

Tato studie se skládá ze dvou experimentálních a jedné dotazníkové části. Detailní instrukce Vám budou zobrazeny na začátku každé části. Délka studie je zhruba 30 minut. Za průběh studie odpovídá Vojtěch Zíka, Ph.D. (vojtech.zika@ujep.cz).

Odměny

Za účast v této studii obdržíte finanční odměnu, která se skládá z fixní a variabilní složky.

Fixní složka odměny. Jako účastník této studie máte nárok na odměnu ve výši 150 Kč.

Tato částka nezávisí na Vašem rozhodování v průběhu studie.

Variabilní složka odměny. Výše a výpočet variabilní odměny bude detailně popsán v každé části. Variabilní odměny jsou v rámci studie uváděny v experimentální měně ECU.

Konverzní poměr mezi ECU a českými korunami je $1 \text{ ECU} = 1 \text{ Kč}$. Variabilní odměna závisí na Vašem rozhodování v průběhu studie. Čtete proto prosím pozorně instrukce a řiďte se jimi; nepozorné nebo náhodné odpovídání může vést ke snížení variabilní složky odměny. Odměna Vám bude vyplacena po skončení studie podle instrukcí administrátora.

Informovaný souhlas

Abyste mohli pokračovat ve studii, potřebujeme od Vás získat informovaný souhlas s účastí ve studii. Věnujte prosím pozornost následujícím informacím.

Účel studie a anonymita dat. Tato studie je pořádána za účelem sběru dat pro akademický výzkum v oblasti lidského chování a rozhodování. Sebraná data jsou anonymní a nelze je zpětně spojit s konkrétní osobou. Data mohou být využita pro sepsání výzkumného článku, který může být publikován v akademickém časopise. V souladu s principy otevřené vědy mohou být anonymizovaná surová data zpřístupněna online na platformách k tomu určených (například osf.io). Zveřejněná data nebudou obsahovat žádné informace umožňující identifikaci účastníků.

Obsah a náročnost studie. Studie se skládá z ekonomických her a dotazníků. Náročnost jakékoliv části této studie nepřekračuje fyzickou nebo psychickou náročnost běžné kancelářské práce.

Chování během studie. Během studie nekomunikujte s ostatními účastníky a řiďte se pokyny administrátorů. Pokud to není nutné, během studie nejezte a nepijte. Všechna elektronická zařízení prosím uveďte do režimu, ve kterém nebudou rušit Vás ani ostatní účastníky. V případě jakýchkoliv otázek či technických problémů zvedněte ruku a v tichosti vyčkejte příchodu administrátora.

Dobrovolnost a předčasné ukončení studie. Účast ve studii je dobrovolná a můžete ji kdykoliv ukončit bez udání důvodů. Pokud budete chtít studii ukončit předčasně, zvedněte ruku a vyčkejte příchodu administrátora. V případě předčasného ukončení studie máte nárok na fixní odměnu za účast na studii, nikoliv však na odměnu získanou během studie samotné.

Souhlas s účastí ve studii. Souhlasíte s podmínkami účasti ve studii? Pokračováním potvrzujete, že jste instrukce přečetli a porozuměli jim.

☐ Ano ☐ Ne

F.2.2 Obrazovka 2 — Demografické otázky

Demografické otázky

Následující otázky slouží výhradně k analytickým účelům. Do pole *Věk* uveďte svůj věk v celých letech. Do pole *Pohlaví* uveďte pohlaví určené při narození; pokud tento údaj nechcete uvést, zvolte možnost "Nechci uvést".

Váš věk: _____

Pohlaví: Žena / Muž / Nechci uvést

F.2.3 Obrazovka 3 — Část I. (Instrukce)

Tato část studie se skládá ze 64 kol, ve kterých můžete sázet na hod spravedlivou mincí, u níž Hlava i Orel padá se stejnou pravděpodobností.

V každém kole uvidíte sekvenci šesti předchozích hodů takovou mincí. Tyto sekvence byly vygenerovány předem a slouží jako modelové scénáře.

Kola jsou rovnoměrně rozdělena do 4 bloků oddělených krátkou přestávkou. Symboly H (Hlava) a O (Orel) jsou v každé sekvenci zobrazeny od nejstaršího hodu vlevo po nejnovější hod vpravo.

Rozhodnutí. Rozhodnutí, které v každém kole činíte, se týká sázky na hod, který následuje po zobrazené sekvenci.

Sázení probíhá prostřednictvím jednoduchého rozhraní, které zobrazuje tři tlačítka: *Hlava*, *Orel* a *Nesázet*:

- Pokud si chcete vsadit, zvolíte *Hlavu* / *Orla* a pomocí posuvníku určíte výši sázky na vybranou stranu.
- Pokud si nechcete vsadit, zvolíte *Nesázet* a pokračujete do dalšího kola bez sázky.
- Pořadí tlačítek *Hlava* a *Orel* se mezi koly může měnit.

Rozhodnutí, která v každém kole učiníte, mají vliv na Vaši odměnu z této části studie.

Odměny. Na začátku každého kola máte k dispozici 100 ECU. V každém kole můžete vsadit libovolnou částku na jednu stranu mince (*Hlavu* / *Orla*), nebo se můžete rozhodnout *Nesázet*. Výpočet odměny se v každém kole řídí stejným pravidlem:

- Nevsazená částka Vám zůstane bez ohledu na to, jaká strana mince v daném kole padne.
- Vsazená částka bude při úspěšné sázce vynásobena 2,5krát. Při neúspěšné sázce o vsazenou částku přijdete.

Nevsazená částka Vám tedy vždy zůstává a vsazená částka se buď znásobí, nebo o ni přijdete.

Na konci studie bude vylosováno jedno z absolvovaných 64 kol. Rozhodnutí, které jste ve vylosovaném kole učinili, určí Vaši odměnu z této části studie. Pokud jste si v tomto kole vsadili, bude proveden hod mincí – padne-li Vámi zvolená strana, byla Vaše sázka úspěšná.

Částka získaná ve vylosovaném kole Vám bude vyplacena v poměru $1 \text{ ECU} = 1 \text{ CZK}$.

Vaši odměnu z této části studie tedy určí rozhodnutí pouze z jednoho kola. Protože však předem nevíte, jaké kolo to bude, je optimální se v každém kole rozhodovat tak, jako by právě ono mělo být vyplaceno.

Zkušební kolo a kontrolní otázky. Po kliknutí na tlačítko se zobrazí zkušební kolo, ve kterém uvidíte rozložení rozhodovací obrazovky tak, jak bude vypadat v ostré části studie.

Zkušební kolo slouží pouze k seznámení se s rozhodovací obrazovkou a nemá vliv na výši Vaší odměny. Po jeho dokončení se Vám zobrazí několik kontrolních otázek, které slouží k ověření porozumění klíčovým aspektům této části studie.

Vaše rozhodnutí pro hod mincí následující po zobrazené sekvenci:

X

Y

Nesázet

Kliknutím na posuvník zvolte vyšší sázky (1–100 ECU).

Sázka: 50 ECU / X

Bude-li toto kolo vybráno k vyplacení a padne zvolená strana, obdržíte 175 ECU.
Pokud padne druhá strana, obdržíte 50 ECU.

Pokračovat

F.2.5 *Obrazovka 5 — Část I. (Kontrolní otázky)*

[Zobrazit instrukce]

- 72

- ☐ 0 ECU
- ☐ 100 ECU
- ☐ 250 ECU

2. Pokud si v kole vybraném k vyplacení rozhodnete vsadit a sázka bude úspěšná, Vaše odměna bude:

- ☐ 100 ECU
- ☐ Nevsazená částka
- ☐ 2,5-násobek vsazené částky
- ☐ Nevsazená částka plus 2,5-násobek vsazené částky

3. Sázka je úspěšná, pokud Vámi vybraná strana odpovídá straně, která:

- ☐ je v zobrazené sekvenci nejčastější
- ☐ je v zobrazené sekvenci poslední
- ☐ padne v hodů spravedlivou mincí, který by následoval po zobrazené sekvenci

4. Odměna z této části studie bude určena na základě rozhodnutí:

- ☐ v jednom náhodně vybraném kole
- ☐ ve všech kolech
- ☐ v posledním kole

5. Který symbol (H = Hlava, O = Orel) je v zobrazené sekvenci nejnovější (poslední)?

- ☐ Vpravo
- ☐ Vlevo

F.2.6 Obrazovka 6–72 — Část I. (Sequence Betting Task)

Obrazovky č. 22, 39 a 56 jsou přestávky mezi jednotlivými bloky:

Dokončili jste 1. ze 4 bloků.

Za 15 sekund budete pokračovat.

F.2.7 Obrazovka 73 — Část II. (Holt-Laury Task)

Vaším úkolem v této části studie je učinit 19 rozhodnutí mezi Možností A a Možností B v tabulce níže.

Odměna z této části bude určena ve dvou krocích. Nejprve bude vylosováno jedno z Vašich 19 rozhodnutí. Poté se u Vámi vybrané možnosti v tomto rozhodnutí určí, zda obdržíte vyšší nebo nižší odměnu, a to následovně:

- Vyšší odměnu (20 ECU u Možnosti A a 38,5 ECU u Možnosti B) získáte s uvedenou pravděpodobností, která se liší v jednotlivých rozhodnutích (např. v prvním rozhodnutí je u obou možností 5 %, ve druhém 10 % atd.).
- Pokud vyšší odměnu nezískáte, obdržíte nižší odměnu (16 ECU u Možnosti A a 1 ECU u Možnosti B).

Takto určená odměna Vám bude vyplacena v poměru $1 \text{ ECU} = 1 \text{ CZK}$.

Shrnutí: Udělejte 19 rozhodnutí mezi Možností A a Možností B. V každém rozhodnutí zvolte možnost, kterou preferujete vzhledem ke kombinaci výše odměny a pravděpodobnosti jejího získání.

Možnost A	Možnost B
20 ECU s pravděpodobností 5 %, jinak 16 ECU	38,5 ECU s pravděpodobností 5 %, jinak 1 ECU
20 ECU s pravděpodobností 10 %, jinak 16 ECU	38,5 ECU s pravděpodobností 10 %, jinak 1 ECU
20 ECU s pravděpodobností 15 %, jinak 16 ECU	38,5 ECU s pravděpodobností 15 %, jinak 1 ECU
20 ECU s pravděpodobností 20 %, jinak 16 ECU	38,5 ECU s pravděpodobností 20 %, jinak 1 ECU
20 ECU s pravděpodobností 25 %, jinak 16 ECU	38,5 ECU s pravděpodobností 25 %, jinak 1 ECU
20 ECU s pravděpodobností 30 %, jinak 16 ECU	38,5 ECU s pravděpodobností 30 %, jinak 1 ECU
20 ECU s pravděpodobností 35 %, jinak 16 ECU	38,5 ECU s pravděpodobností 35 %, jinak 1 ECU
20 ECU s pravděpodobností 40 %, jinak 16 ECU	38,5 ECU s pravděpodobností 40 %, jinak 1 ECU
20 ECU s pravděpodobností 45 %, jinak 16 ECU	38,5 ECU s pravděpodobností 45 %, jinak 1 ECU
20 ECU s pravděpodobností 50 %, jinak 16 ECU	38,5 ECU s pravděpodobností 50 %, jinak 1 ECU
20 ECU s pravděpodobností 55 %, jinak 16 ECU	38,5 ECU s pravděpodobností 55 %, jinak 1 ECU
20 ECU s pravděpodobností 60 %, jinak 16 ECU	38,5 ECU s pravděpodobností 60 %, jinak 1 ECU
20 ECU s pravděpodobností 65 %, jinak 16 ECU	38,5 ECU s pravděpodobností 65 %, jinak 1 ECU
20 ECU s pravděpodobností 70 %, jinak 16 ECU	38,5 ECU s pravděpodobností 70 %, jinak 1 ECU
20 ECU s pravděpodobností 75 %, jinak 16 ECU	38,5 ECU s pravděpodobností 75 %, jinak 1 ECU
20 ECU s pravděpodobností 80 %, jinak 16 ECU	38,5 ECU s pravděpodobností 80 %, jinak 1 ECU
20 ECU s pravděpodobností 85 %, jinak 16 ECU	38,5 ECU s pravděpodobností 85 %, jinak 1 ECU
20 ECU s pravděpodobností 90 %, jinak 16 ECU	38,5 ECU s pravděpodobností 90 %, jinak 1 ECU
20 ECU s pravděpodobností 95 %, jinak 16 ECU	38,5 ECU s pravděpodobností 95 %, jinak 1 ECU

F.2.8 Obrazovka 74 — Část III. (Life Orientation Test–Revised)

Odpovězte prosím na následující otázky týkající se Vaší osoby. Míru svého souhlasu uveďte pomocí níže uvedené škály (1 = rozhodně nesouhlasím; 5 = rozhodně souhlasím).

1. V nejistých dobách obvykle očekávám to nejlepší.
2. Je pro mě snadné se uvolnit.
3. Pokud se v mém životě něco může pokazit, tak se to pokazí.
4. Na svou budoucnost se vždy dívám optimisticky.
5. Velmi si užívám společnost svých přátel.
6. Je pro mě důležité neustále něco dělat.
7. Téměř nikdy neočekávám, že věci půjdou tak, jak bych si přál/a.
8. Nenechám se snadno rozhodit.
9. Málokdy počítám s tím, že se mi přihodí něco dobrého.
10. Celkově očekávám, že se mi v životě přihodí více dobrého než špatného.

(Pod každou položkou byla zobrazena tato škála:)

- ☐ rozhodně nesouhlasím
- ☐ nesouhlasím
- ☐ ani nesouhlasím, ani souhlasím
- ☐ souhlasím
- ☐ rozhodně souhlasím

F.2.9 Obrazovka 75 — Odměna

Vaše odměna

Fixní odměna za účast:	150,00 Kč
Variabilní odměna:	219,00 Kč
Hra se sekvencemi: k výplatě bylo vylosováno kolo 36, získaná odměna	180,00 Kč
Výběr loterií: k výplatě byl vylosován řádek 17, získaná odměna	39,00 Kč
CELKOVÁ ODMĚNA:	369,00 Kč

F.2.10 Obrazovka 76 — Závěrečné otázky

Následující otázky nám pomohou lépe porozumět Vašemu rozhodování během sázení na hod mincí. Odpovědi nemají vliv na výši Vaší odměny.

1. Pokud byste v úloze se sekvencemi hráli 100 kol, v kolika kolech byste podle Vás vyhráli? (0 = v žádném kole; 50 = v polovině kol; 100 = v každém kole)
2. Jak riskantní Vám přišlo sázení v úloze se sekvencemi? (0 = vůbec; 10 = velmi)
3. Do jaké míry jste věřil(a), že každý hod má pravděpodobnost 50 % na Hlavu a 50 % na Orla bez ohledu na zobrazenou historii šesti hodů? (0 = vůbec ne; 10 = zcela)
4. Do jaké míry jste při rozhodování zohledňoval(a) zobrazenou historii šesti hodů, i když jste věděl(a), že mince je spravedlivá? (0 = vůbec ne; 10 = velmi silně)
5. Jak často jste sázel(a) hlavně proto, že Vám přišlo lepší "něco udělat" než nesázet? (0 = nikdy; 10 = velmi často)
6. Nakolik jste se během úlohy nudil(a) nebo byl(a) unavený(á)? (0 = vůbec; 10 = velmi)
7. Jak byste celkově popsal(a) svou ochotu podstupovat finanční riziko? (0 = velmi nerad(a) riskuji; 10 = velmi rád(a) riskuji)
8. Používal(a) jste při rozhodování nějaké pravidlo nebo strategii?

☐ Ano

☐ Ne

☐ Nejsem si jistý(á)

9. Pokud ano, stručně ji popište: (*volitelné*)

10. Máte k této části studie jakýkoli další komentář? (*volitelné*)

F.2.11 Obrazovka 77 — Konce studie

Děkujeme za účast!

Odměna za tuto studii Vám bude vyplacena bankovním převodem. Nyní se prosím ujistěte, že:

- Máte připravený QR kód pro platbu. Částku doplní administrátor. Pokud nevíte, jak na to, připravte si číslo svého účtu.
- Máte míček, který jste si vylosovali při vstupu do laboratoře a který určuje číslo Vašeho sedadla.

Máte-li připravený QR kód (nebo číslo účtu) a držíte-li v ruce míček, přihlaste se a vyčkejte na instrukce administrátora studie.

G TRANSLATION OF LIFE ORIENTATION TEST–REVISED

We translated the Life Orientation Test–Revised (Scheier, Carver, and Bridges 1994) from English to Czech following a forward-backward translation method to ensure linguistic equivalence (Brislin 1976). The translation was carried out by Spěvák, a professional Czech translation agency. In the first phase, two independent translators translated the scale from the original English into Czech. In the second phase, a third independent translator reconciled these two versions into a single Czech version. A native English speaker then translated this combined version back into English. The back-translated version was compared to the original, differences in content and meaning were identified, and the Czech version was adjusted accordingly. This process yielded the final Czech version of the scale reproduced below.

Test Scoring:

1. Reverse code items 3, 7, and 9 prior to scoring (0=4) (1=3) (2=2) (3=1) (4=0).
2. Sum items 1, 3, 4, 7, 9, and 10 to obtain an overall score.

Note: Items 2, 5, 6, and 8 are filler items only. They are not scored as part of the revised scale.

G.1 English Original

Revised Life Orientation Test (LOT-R)

Instructions: Please answer the following questions about yourself by indicating the extent of your agreement using the following scale:

[0] strongly disagree

[1] disagree

[2] neutral

[3] agree

[4] strongly agree

Be as honest as you can throughout, and try not to let your responses to one question influence your response to other questions. There are no right or wrong answers.

- _____ 1. In uncertain times, I usually expect the best.
- _____ 2. It's easy for me to relax.
- _____ 3. If something can go wrong for me, it will.
- _____ 4. I'm always optimistic about my future.
- _____ 5. I enjoy my friends a lot.
- _____ 6. It's important for me to keep busy.
- _____ 7. I hardly ever expect things to go my way.
- _____ 8. I don't get upset too easily.
- _____ 9. I rarely count on good things happening to me.
- _____ 10. Overall, I expect more good things to happen to me than bad.

G.2 Czech Translation

Revidovaný test životní orientace (LOT-R)

Pokyny: Odpovězte prosím na následující otázky týkající se vaší osoby – míru svého souhlasu uveďte pomocí následující škály:

- [0] rozhodně nesouhlasím
- [1] nesouhlasím
- [2] ani nesouhlasím, ani souhlasím
- [3] souhlasím
- [4] rozhodně souhlasím

Buďte co nejupřímnější a snažte se, aby odpověď na jednu otázku neovlivnila odpovědi na jiné otázky. Neexistují správné ani špatné odpovědi.

- _____ 1. V nejistých dobách obvykle očekávám to nejlepší.
- _____ 2. Je pro mě snadné se uvolnit.
- _____ 3. Pokud se v mém životě něco může pokazit, tak se to pokazí.
- _____ 4. Na svou budoucnost se vždy dívám optimisticky.
- _____ 5. Velmi si užívám společnost svých přátel.
- _____ 6. Je pro mě důležité neustále něco dělat.
- _____ 7. Téměř nikdy neočekávám, že věci půjdou tak, jak bych si přál/a.
- _____ 8. Nenechám se snadno rozhodit.
- _____ 9. Málokdy počítám s tím, že se mi přihodí něco dobrého.
- _____ 10. Celkově očekávám, že se mi v životě přihodí více dobrého než špatného.

H DEPARTURES FROM THE PREREGISTRATION

confirmatory sample no in class experiment

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